

The development of stratification and segregation in a new scientific field: A study of collaboration among scientists in neuroblastoma research between 1975 and 2016[☆]

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ABSTRACT

Using a new data set on scientific collaboration in neuroblastoma research over a period of 41 years, we study how the structure of collaboration ties in an evolving scientific field changes over time. Guided by concepts from the sociology of knowledge and status theories, we highlight the importance of a field's size and age for processes of stratification and segregation within it. Our findings suggest that stratification becomes stronger and diversifies as the field of neuroblastoma research expands. Moreover, we find that the occurrence of collaboration between researchers with a similar status becomes more likely as the field matures. While previous work has primarily examined established fields, our study focuses on how scientific fields change their structure over time. More broadly, our article demonstrates the potential analytical merits of adopting a dynamic and relational perspective on the developmental trajectories of organizational and cultural fields.

1. Introduction

Both previous and recent research shows that accumulation dynamics shape the structure of scientific collaboration and perpetuate inequality in the distribution of recognition and resources within scientific communities (Allison et al., 1982; Blau, 1994; Bol et al., 2018; Bourdieu, 1988; Burris, 2004; Cole and Cole, 1973; Eom and Fortunato, 2011; Gondal, 2018; Hagstrom, 1971; Lynn, 2014; Merton, 1968).

We contribute to this line of work by studying how stratification and segregation shape scientific collaboration networks throughout the evolution of a new scientific field. The empirical setting of our study is a specialized community of cancer researchers who began to investigate neuroblastoma—the most common solid cancer in childhood (Maris, 2010)—in the second half of the 1970s. Consequently, our theoretical framework draws on insights from the sociology of knowledge (Chubin, 1976; Cole and Cole, 1973; Crane, 1972; Mullins, 1972), status theories (Gould, 2002; Podolny, 2010; Ridgeway and Correll, 2006), and a growing body of research that applies network analysis to study the

structure of coauthor or citation networks (Dahlander and McFarland, 2013; Ferligoj et al., 2015; Foster et al., 2015; Friedkin, 1978; Gondal, 2011; Kronegger et al., 2012; Lynn, 2014; Ma et al., 2020; Moody, 2004; Shwed and Bearman, 2010; Stark et al., 2020).

Sociologists have identified the Matthew effect—a tendency of actors with a large stock of resources and recognition to accumulate even more of these assets—as an important factor for the social organization of scientific communities (Cole and Cole, 1973; Crane, 1972; Merton, 1988, 1968). Thereby, preferential attachment can be regarded as a special case of the Matthew effect and describes a process by which actors with many network ties tend to attract more ties over time (Barabási et al., 2002; Barabási and Albert, 1999). While scientometric, sociological, and network-scientific studies persistently report a concentration of coauthorships among an elite of scholars (Barabási et al., 2002; Leydesdorff and Wagner, 2008; Moody, 2004; Newman, 2001a), less systematic attention has been paid to the question of how inequality in the distribution of network partners changes as scientific fields mature.

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Furthermore, although canonical works suggest that scientific fields exhibit different forms of social organization at the outset compared to during their later stages (Chubin 1976; Crane 1972: 85–98, 115–128; Mullins 1972), no network-analytical study to date has examined how accumulation dynamics operate through multiple channels throughout a network's evolution.

Moreover, our investigation of the structure of a new scientific field that has gradually transformed over more than 40 years also contributes to a stream of literature that emphasizes the role of contextual characteristics for network structure (Martin, 2009; McFarland et al., 2014; Simpson, 2019; White, 2008). As McFarland et al. (2014: 1112) note, not only do the preconditions for processes of tie formation vary between contexts, they also change over time. Yet, thus far only a few sociological studies have examined changes in network mechanisms as a context matures (Lewis and Kaufman, 2018; Schaefer and Kreager, 2020). We address this research gap by studying changes in the structure of collaboration networks in a new scientific field.

To guide our analysis, we build upon the assumption that members of a nascent scientific specialty develop their careers by mobilizing new members (Chubin, 1976; Frickel and Gross, 2005; Mullins, 1972). Established actors thereby acquire more resources as a field gains legitimacy and in turn receives more external funding (Alberts et al., 2014; Laudel, 2006). Consequently, senior scientists are able to integrate new researchers into their teams and strengthen their position in the community (Bourdieu, 1988; Frickel and Gross, 2005; Lazega et al., 2006; Mullins, 1972).

Moreover, status theories suggest that status recognition affects collaboration dynamics in organizational fields and helps actors to navigate complex social environments (Lynn, 2014; Podolny, 2010; Sauder et al., 2012). Similarly, status construction theory highlights that status beliefs reduce situational uncertainty and ease coordination problems (Ridgeway, 1991; Ridgeway and Correll, 2006; Ridgeway and Erickson, 2000). Following these literatures, we argue that the presence of a larger number of authors in a field increases uncertainty, which elevates the importance of status recognition for coauthor choices (cf., McFarland et al., 2014).

Both streams of literature lead us to the expectation that inequality in the distribution of coauthorships increases throughout the evolution of scientific fields. Furthermore, we propose that a diversification of accumulation dynamics accompanies this trend: as a field grows, we expect years of experience, productivity, and seniority to factor into scientists' popularity as coauthors. In addition, we hypothesize that status homogeneity—the tendency to collaborate with others similar in status—increasingly structures scientific collaboration in later stages of a field. At the same time, mentor-apprentice ties between early career researchers and established scientists should retain their relevance for scientific collaboration.

To test our theoretical arguments, we apply exponential random graph models (ERGMs; Lusher et al., 2013) to a coauthor network based on abstracts submitted to the Advances in Neuroblastoma Research (ANR) conferences between 1975 and 2016 (Berthold et al., 2019). This manually collected data source is well suited for studying how a scientific specialty evolved because, initially, neuroblastoma was a new topic addressed by only a small number of researchers (Martynov et al., 2020). While previous studies examined entire scientific disciplines (Moody, 2004; Newman, 2001a), national research communities (Ferligoj et al., 2015; Kronegger et al., 2012, 2011; Lazega et al., 2006), or collaboration within a single university (Dahlander and McFarland, 2013; Rawlings et al., 2015; Stark et al., 2020), we focus on the long-term trajectory of a scientific field organized around a demarcated, well-defined research topic that was not diluted by other topics throughout the four decades under observation.

Our findings indicate that the concentration of collaboration on an elite of researchers becomes stronger as neuroblastoma research matures. Also, we find that collaboration is increasingly segregated according to researchers' productivity and experience over time.

2. Theory

Social scientists have firmly established the belief that social processes affect knowledge production (Bourdieu, 1988; Chubin, 1976; Crane, 1972; Frickel and Gross, 2005; Knorr, 1999; Latour, 1987; Lazega et al., 2006; Merton, 1988, 1968; Mullins, 1972). Scientific fields resemble *invisible colleges* (Crane, 1972) marked by interorganizational collaboration and regular communication among researchers. Likewise, scientific specialization revolves around clusters of researchers connected by workshops, conferences, and informal social meetings (Chubin, 1976; Mullins, 1972). Moreover, the growing size of research teams—for instance, in high energy physics (Newman, 2001b, 2001c)—makes modern science an increasingly social activity (Leahey, 2016; Wuchty et al., 2007).

Concerning stratification in science, Merton (1968) noted the self-reinforcing nature of accumulation dynamics many years ago and coined the term “Matthew effect”, which describes that well-known scholars gain more resources and recognition over time by benefitting from initially small relative advantages. For the case of network ties, this claim has since been supported by many studies analyzing coauthorships or citations and is usually called “preferential attachment” in network-analytical applications (Barabási et al., 2002; Eom and Fortunato, 2011; Newman, 2001a). While an accumulation of coauthorships by an elite of researchers is often viewed as an average tendency—persistently shaping the structure of scientific collaboration—contemporary and earlier theoretical accounts suggest that the structure of organizational (Hannan and Freeman, 1993; Padgett and Powell, 2012), cultural (Baumann, 2001; Becker, 2008; Bourdieu, 1993, 1984; White and White, 1993), and scientific fields (Chubin, 1976; Crane, 1972; Frickel and Gross, 2005; Jurgens et al., 2018; Kuhn, 1970; Munoz-Najar Galvez et al., 2019) changes over time. In the present study, we elaborate on these dynamic perspectives on the structural configurations of various fields and derive theoretical expectations for the developmental trajectory of a developing scientific specialty in the next section.

2.1. Inequality in a developing scientific field

Our first theoretical expectation concerns the question of whether a new scientific field exhibits different levels of stratification throughout its evolution. Previous work suggests that scientists first have to allocate resources to make new epistemic claims that are credited by a peer audience (Boardman and Ponomarev, 2007; Chubin, 1976; Frickel and Gross, 2005). Also, scientists need to mobilize collaborators such as PhD students and postdocs to generate a high output of well-cited publications, thus allowing them to acquire more resources and to stabilize their position in a new field of inquiry (Griffith and Mullins, 1972; Latour and Woolgar, 1986; Lazega et al., 2016, 2006; Li et al., 2013).¹

In terms of the structure of scientific collaboration, these processes produce “hub-spoke structures” (Martin, 2009), with senior scientists acting as hubs and younger researchers as spokes. These constellations induce stratification in the number of publications since the leader of a research group is usually named as a coauthor on all the papers that a

¹ While we draw upon empirical findings stemming from the stream of science and technology studies (STS), we would like to point out that the usage of network-theoretic concepts in this tradition differs from our focus on collaborative networks (Venturini et al., 2019). According to the STS perspective, networks can be composed of entities such as documents, laboratories, and other parts of scientific infrastructure as well as human persons (Foster et al., 2015; Knorr, 1999; Latour, 1987; Shi et al., 2015). In comparison, sociologists of science and researchers using social network analysis tend to study social networks among persons—e.g., networks of collaborations or citations among scholars (Chubin, 1976; Crane, 1972; Dahlander and McFarland, 2013; Gondal, 2011; Mullins, 1972; Rawlings et al., 2015; Stark et al., 2020).

group produces. In contrast, PhDs and postdocs tend to work on a small set of papers as primary investigators (see, for example, Knorr, 1999: Ch. 9). Moreover, leaders often function as representatives of their group—for instance, by promoting the group at conferences or by contacting leaders of other research groups (Griffith and Mullins, 1972: 961; Knorr, 1999: 222–224; Lazega et al., 2016, 2006). This role allows them to forge collaborations between groups, which increases leaders' privileged access to coauthor ties (see, for example Hancean et al., 2021).

Regarding the global trajectory of a field, we argue that the overall inequality in collaboration is likely to increase as an area of scientific inquiry grows. This should be the case because collaborations between research groups—often initiated by senior scientists (Latour and Woolgar, 1986)—increase in maturing scientific specialties (e.g., Mullins, 1972), and the structure of laboratories changes due to new funding sources (Alberts et al., 2014; Laudel, 2006). If new entrants collaborate in the same way as previous field members, inequality should remain stable. Yet, we expect that the balance between scientists with tenured positions and early career researchers shifts as a field grows.

As more external funding is channeled into a new field of inquiry, funding agencies mainly offer programs that provide short-term employment for PhDs and postdocs. In contrast, the number of tenured positions does not increase proportionally because universities and research facilities are reluctant to create costly long-term positions (Laudel, 2006; Münch, 2014). As Alberts et al. (2014) point out in regard to US biomedical research: although the field experienced rapid growth from the 1980s onward, the career prospects of early career researchers worsened decisively as the influx of new funding “has led to an enormous growth in ‘soft money’ positions, with stagnation in the ranks of faculty who have institutional support” (Alberts et al., 2014: 5775). Consequently, we expect that established scientists employ a larger staff of early career researchers if more external funding opportunities emerge and that collaborations will concentrate on a smaller proportion of the field. Thus, in turn, we expect a nascent scientific field to exhibit less inequality in the distribution of coauthorships than a mature field.

This expectation can also be derived from another stream of literature concerned with the ubiquity of status processes in markets, organizational fields, and interaction in task-oriented groups (Borkenhagen and Martin, 2018; Gondal, 2018; Ma et al., 2020; Podolny, 2010; Ridgeway, 1991, 2019; Sauder et al., 2012). What is common to different usages of status as a theoretical concept is a definition of status as prestige, respect, recognition, and (psychological) deference received by others (Fiske, 2011; Gould, 2002; Podolny, 2010; Ridgeway, 2019).

In line with Podolny (2010), we assume that actors use status signals as cues to reduce uncertainty, thus allowing them to navigate complex fields.² Likewise, status construction theory suggests that status beliefs emerge from a concatenation of micro-interactions due to actors' need to reduce situational uncertainty. In particular, status beliefs allow actors to solve the problem of coordinating behavior, which is necessary to achieve group goals (e.g., Ridgeway and Correll, 2006: 6). As uncertainty increases, status should play a more prominent role in actors' behavior because the heuristic usefulness of status categorizations increases as environments or situations become more complex (Blau, 1968; Fiske, 2011; Mayhew, 1973; McFarland et al., 2014; Podolny, 2010; Ridgeway, 2019).

Regarding cooperation in science, most fields initially consist of a small set of actors. Therefore, it is still feasible to monitor the actions of others in detail. As a field matures, new actors enter and it becomes more difficult, or even impossible, to keep track of what others are doing. To

retain their capacity to navigate the field—in other words, to decide whom they should collaborate with or whom they should cite—actors tend to apply filters to the relational information they receive (Brashears, 2013; Brashears and Quintane, 2015; Lynn, 2014; Mayhew, 1973; Mayhew and Levinger, 1976).

Consequently, we expect more inequality in the distribution of coauthorships over time because a growing field increases uncertainty, which in turn amplifies the influence of status recognition on the formation of coauthor ties.

Hypothesis 1. Inequality in the distribution of coauthorship ties increases as a scientific field matures and grows.

In summary, we expect rising inequality in the distribution of coauthorships due to a shrinking proportion of actors who manage to mobilize additional researchers and resources (Alberts et al., 2014; Chubin, 1976; Frickel and Gross, 2005; Laudel, 2006; Mullins, 1972), and due to the increasing importance of status as a cognitive heuristic shaping social interaction in uncertain environments (Podolny, 2010; Ridgeway and Erickson, 2000).

2.2. Diversification of accumulation dynamics

In addition to a trend towards more stratification in coauthor ties, we also consider different channels through which actors manage to attract coauthors. A scientist's prominence as a collaborator might result from their ability to offer others resources to conduct studies, a reputation for technical expertise in a particular research area, the skill to spark interest for new topics, or experience in writing academic papers and applying for funding (Blau, 1994; Dahlander and McFarland, 2013; Griffith and Mullins, 1972; Knorr, 1999; Lamont, 2009; Merton, 1968; Mullins, 1972; Newman, 2001a; Zuckerman, 1968). Regarding the question of how multiple aspects structure the distribution of coauthorships, we expect accumulation dynamics tied to years of experience, productivity, and seniority to diversify throughout the evolution of a new scientific field.

Following accounts that stress the temporal unfolding of research communities, we expect that accumulation dynamics diversify because as actors develop their careers, they can draw upon multiple field-specific resources to mobilize additional collaborators (Chubin, 1976; Cole and Harriet, 1975; Frickel and Gross, 2005; Griffith and Mullins, 1972). In the early stages of a scientific endeavor, pioneering researchers primarily rely on their charisma and their ability to spark interest in new topics. In a comparison of different specialties, Griffith and Mullins (1972: 961) highlight the role of leaders who are crucial in organizing the intellectual activities of a nascent specialty but who initially often lack tangible resources such as funding.³

Consequently, building and maintaining collaborations despite these unfavorable conditions should be the prime factor influencing researchers' prominence as coauthors during the genesis of a field.⁴ Throughout its development, a field gains legitimacy, new external funding sources become available, and successful leaders of research groups forge their

² The notion that actors use cognitive heuristics to store and represent relational information is well supported by empirical evidence (Brashears, 2013; Carnabuci et al., 2018; Krackhardt and Kilduff, 1999). Perceiving others on a vertical dimension plays a crucial role in social cognition and shapes interaction across a variety of settings (Anderson, Hildreth, and Howland, 2015; Berger, Cohen, and Zelditch, 1972; Fiske, 2011; McMahan, 2017; Ridgeway, 1991).

³ Furthermore, resources acquired in adjacent fields—e.g., previous academic positions or publications—are seldom sufficient to attract aspiring researchers because unorthodox scientific endeavors present much higher risks to the success of individuals' careers compared with work on established problems (Frickel and Gross, 2005; Griffith and Mullins, 1972; Latour and Woolgar, 1986). As Latour (1987) observed, scientists who tackle new problems often face severe resistance from established scientific elites or an inadequate infrastructure for their research (see also Bourdieu, 1988; Frickel and Gross, 2005).

⁴ For instance, Mullins (1972) showed that the scientific specialty of phage work—which led to the new discipline of molecular biology—was initially driven by a small circle of charismatic leaders who managed to recruit students through informal social gatherings and workshops for their cause. Scientists only received more institutional support and started to build a stock of influential publications as phage work matured (Mullins, 1972: 74).

careers (Alberts et al., 2014; Laudel, 2006). Thus, we expect field-specific resources such as years of experience, productivity, and seniority to gain importance for the acquisition of coauthorships over time.

Again, we can also build on status theories to arrive at this expectation (Gould, 2002; Lynn et al., 2009; Podolny, 2010; Ridgeway, 2019; Ridgeway and Correll, 2006). Ridgeway (2019) highlights that status characteristics influence interaction in task-oriented groups. Similarly, Podolny (2010) proposes that status signals guide collaborator choices in organizational fields. Following these accounts, we argue that the publications and years of experience a researcher has accumulated start to act as status markers as a field matures. As outlined by Ridgeway (2019), initially, the ascription of status to nominal characteristics, such as gender, is arbitrary and does not necessarily correspond with actors' contributions to group goals or their competence. Yet, even small differences in the ascription of status that align with nominal characteristics can initiate a self-reinforcing dynamic whereby status beliefs spread and lead to persistent inequality in recognition and resources on the macro level (Grow et al., 2015; Mark et al., 2009; Ridgeway, 1991, 2014).

In our case, we expect that the ascription of status is arbitrary at the beginning of a field but that a growing set of status markers crystalizes over time because researchers have to navigate an increasingly complex environment (Blau, 1968; Fiske, 2011; Mayhew, 1973; Podolny, 2010). While knowing, for instance, how many coauthors a scientist has worked with provides sufficient information to judge their status in the early stages of a new field, additional status markers such as years of experience, publications, or seniority should start to inform collaborator choices at later developmental stages.

Hypothesis 2. Accumulation dynamics diversify as a scientific field matures and grows.

2.3. Closure according to status differences, a consolidating elite

Besides more intense and diverse stratification, we also consider how status similarity between researchers affects collaboration during different periods of a field. Previous research suggests that actors' close relationships or coalitions are segregated along the lines of status—for example, adolescents tend to form friendship groups of similar-status individuals (Adler and Adler, 1998; Coleman, 1961; Milner, 2013) and organizations sharing a similar status are more likely to collaborate (Podolny 2010). Additionally, scholars who studied status orders among university departments found that elite departments are more likely to exchange PhD students and to collaborate with other elite departments (Burris, 2004; Gondal, 2018; Han, 2003; Ma et al., 2020).

Before increasing closure according to status differences emerges, we expect that researchers working in a nascent field primarily collaborate with others who are dissimilar in years of experience, productivity, and seniority. This tendency toward *status heterogeneity* should follow from the organizational logic of local research activities, which tend to be divided between experienced scholars and apprentices (Blau, 1994; Knorr, 1999; Latour, 1987; Latour and Woolgar, 1986; Lazega et al., 2006). Senior scientists offer new opportunities to conduct research within departments and primarily collaborate with PhD candidates and postdocs (e.g., Mullins, 1972). Thus, scientists who hold many collaboration ties should work with others who hold fewer ties, especially if more external funding becomes available in a field and creates a larger scientific staff for leaders of established laboratories (Alberts et al., 2014; Laudel, 2006). While these instances of local, status-dissimilar collaboration should not lose their significance over time, we expect more collaboration between status-similar senior scientists as a field matures according to status markers—i.e., increasing *status homogeneity*.

As Knorr (1999: 235–240) shows, leaders of research groups act as representatives and forge collaborations with other groups working on the same topics (see also, Chubin, 1976; Griffith and Mullins, 1972; Mullins, 1972). Yet, this is only possible if a momentary consensus about relevant research areas and methods crystallizes (Koppman and Leahey,

2016; Latour, 1987; Lazega et al., 2016; Schwemmer and Wieczorek, 2020; Shi et al., 2015; Shwed and Bearman, 2010). Therefore, we expect more cases of status-similar collaboration as a field matures and senior scientists begin to form coauthor ties outside their local environments with leaders of other groups pursuing similar research.

In addition, we expect the formation of circles of authors who are status-similar due to increased status homophily in uncertain contexts.⁵ In line with Podolny (2010), we argue that this stems from high-status actors' fear of being associated with low-status actors. As Podolny (2010, 24–39) highlighted, status “leaks” through social relationships, whereby the status of actors' collaborators rubs off on their own status. Corroborating this theoretical expectation, Podolny (2010, 76–102) finds that status homogeneity can be observed in the investment banking industry, particularly under greater uncertainty. Following this research, we argue that collaboration with low-status scientists sends negative signals within an academic community and that the balance between status-dissimilar and status-similar collaborations tends to shift in favor of status-similar collaborations as the field matures.

Hypothesis 3. Scientists who are dissimilar in their years of experience, productivity, and seniority collaborate at the outset of a field. This tendency weakens in the later stages of a field's development, and collaboration increasingly exhibits status homogeneity.

Overall, our theoretical expectations sketch a trajectory marked by the growing importance of accumulation dynamics for the structure of scientific collaboration. We argue that an elite of authors emerges as a field matures, based on the accumulation of coauthorship through multiple channels such as years of experience, productivity, and seniority. Also, senior scientists collaborate with status-similar researchers and their apprentices as a field ages.

3. The emerging field of neuroblastoma research

To test our theoretical expectations, we focus on the development of a specialized scientific community devoted to neuroblastoma research. Neuroblastoma represents the most common solid cancer in childhood (Maris, 2010). A special characteristic of neuroblastoma is the vast diversity of possible tumor types and consequences for patients: while most infants with neuroblastoma may recover entirely with minimal treatment⁶, children who are older than one year often die or face long-term health conditions due to high-dose chemotherapy, radiation, and immunotherapy (Brodeur, 2003; Cheung and Heller, 1991; Matthay et al., 1999).

Neuroblastoma is a malignant tumor of the sympathetic autonomic nervous system and was first described by R. Virchow in 1864 as a neoplasm originating from the organ adrenal medulla.⁷ While research continued into the late 19th and early 20th centuries (Hutchison, 1907; Pepper, 1901)⁸, scientists only started to develop a clearer consensus on

⁵ Please note that we use the term “status homogeneity” instead of “status homophily” because “homophily” has strong connotations of a social-psychological preference for similar others (McPherson et al., 2001). As Wimmer and Lewis (2010) point out, network segregation can originate in different sources such as the opportunity structure (Blau, 1977) or endogenous network mechanisms. Here, we remain agnostic as to the question of what drives status homogeneity in scientific networks and focus on changes over time. Furthermore, the methods we use cannot distinguish between influence and selection (Shalizi and Thomas, 2011; Steglich et al., 2010). Therefore, we cannot determine whether actors of similar status select each other as collaborators or whether connections to high-status alters are elevating actors' status.

⁶ Merely tumor resection and observation of the primary tumor sites (e.g., Maris, 2010).

⁷ The adrenal medulla is part of the adrenal glands, which lie above the kidneys and produce various hormones (e.g., Avisse et al., 2000).

⁸ Marchand (1891) disclosed the common features of the sympathetic nervous system and the adrenal medulla. Pepper (1901) and Hutchinson (1907) described different biological prototypes.

how to classify and study neuroblastoma during the 1980s due to advances in the field of histopathology—the microscopic study of diseased biological tissues (Dehner, 1988; Olson, 1989; Shimada et al., 1984). Moreover, the first internationally accepted staging system was proposed by Evans et al. (1971), which paved the way for progress in research, diagnosis, and treatment.

Since then, those involved in researching and treating neuroblastoma have achieved higher survival rates among patients—especially for low- and intermediate-risk neuroblastoma—while also achieving earlier detection of the disease and providing more insights about its potential causes (Brodeur et al., 1993; Maris, 2010). However, high-risk neuroblastoma still leads to very high mortality rates, and the molecular mechanisms underpinning the disease continue to be subject to research efforts (Ray, 2019).

The scholars who have entered this community came from various disciplines such as medicine, biology, and chemistry (Martynov et al., 2020). In the first third of the field's development, clinical researchers and biologists made up the majority of scholars participating in neuroblastoma research. From the 1990s onward, though, the field experienced an influx of bio-molecular and bio-chemical researchers as well as immunologists, which broadened the interdisciplinary scope of the community (Martynov et al., 2020).⁹

4. Data

To study how the social organization of neuroblastoma research changed over time, we analyze information on abstracts submitted to the Advances in Neuroblastoma Research (ANR) conference series. The initial ANR conference was the first interdisciplinary meeting devoted explicitly to neuroblastoma and hosted by the Children's Hospital of Philadelphia in 1975. We consider 18 conferences until 2016, which all addressed neuroblastoma from the perspective of basic, translational, and clinical science without dilution by other topics (Berthold et al., 2019).

The authors digitalized all the abstracts from 18 conferences documented in the conference proceedings books (for details, see Berthold et al., 2019). The resulting data set spans 41 years (1975–2016). We used the co-occurrence in abstracts of papers presented at the ANR to derive collaboration ties between researchers at each conference.

Following the idea that repeated interaction and communication among researchers is crucial for the emergence of a community (Chubin, 1976; Crane, 1972; Mullins, 1972), we decided to only include researchers in our study who appeared as authors of abstracts for at least two conferences (~40% of the initial sample).¹⁰

5. Measures

5.1. Years of experience

To measure the length of scientists' careers, we operationalized experience as years since the first occurrence of an author in our data set. Descriptive statistics on scientists who participated in the conference series at least twice are provided in Appendix A.

⁹ This development was accompanied by literature that studied the genetic characteristics of neuroblastoma and has led to new forms of therapy and a better understanding of the disease on a molecular level (Brodeur, 2003; Kaghad et al., 1997; Mossé et al., 2008).

¹⁰ Focusing the analysis on authors who appeared at least twice in abstracts over the years under observation seemed reasonable to us in light of the general trend in scientific research toward more publications per author and the expansion of research teams (Wuchty et al., 2007). Whereas many authors who appear only once might be included in the paper but not attend the conference or contribute much to the actual paper, the likelihood of capturing a real collaboration between authors should be higher for the stable part of the sample.

5.2. Productivity

We accessed Clarivate's Web of Science database to capture scientists' research output during the period 1975–2017 and merged the obtained publications with our data set on conference participation.¹¹ Because the conference data did not provide full names for every author and some names and abbreviations were ambiguous, we followed a step-wise disambiguation strategy: in the first step, we identified authors who could not be unambiguously connected to publications via abbreviated names (123 out of 8460 authors). In the second step, we manually linked these authors with their publications under consideration of their institutional affiliations. To ensure a correct linkage, an experienced neuroblastoma scientist carried out an in-depth investigation and achieved unambiguous correspondence between authors and publications in nearly all cases.

We restricted our search to articles on the subject "neuroblastoma" and considered the headlines, abstracts, and keywords of published articles. Hence, the search according to subject was preferred over a purely title-based approach (Tal and Gordon, 2017). We pursued this focused search strategy to avoid an overlap with adjacent research fields, as our aim was to capture publications within the scientific specialty of neuroblastoma research. Also, we excluded abstracts, meeting reports, presentations at satellite workshops, and reviews, as these carry different meanings and functions in research communities compared with peer-reviewed articles (e.g., Lamont, 2009). This decision was motivated by our goal to obtain a homogeneous measure for researchers' publication output over a long period.

5.3. Seniority

We measured seniority by constructing a variable that indicates how often a researcher is listed as the last author on one of the papers presented. In a similar vein to the relative position of investment banks in advertisements announcing a new deal (Podolny 2010: 40–76), the positions of author names in publications and conference abstracts in science are unequally prestigious: in particular, the last position in publications is more prestigious than other positions, as this person represents—in many instances—the leader of a research group (Bennett and Taylor, 2003; Costas and Bordons, 2011; Knorr, 1999; Latour and Woolgar, 1986; Savitz, 1999; Shapiro et al., 1994; Zuckerman, 1968).¹²

5.4. Stratification

We use the skewness of the degree distribution of the collaboration network as a global measure for inequality at a given point in time. If a degree distribution is positively skewed, this is indicative of a small number of individuals having many ties, while the majority of individuals exhibit none or only a few ties (Fisher, 2018: 57; Moody et al., 2011: 103). As additional measures for inequality, we repeated our analyses with the standard deviation and the Gini coefficient (Badham, 2013; Snijders and Steglich, 2015). Results for these measures are reported in Appendix A.

¹¹ Accessed July 17, 2017.

¹² Please note that last author positions are not necessarily prestigious in all disciplines or fields of scientific inquiry. Therefore, the measure for seniority we propose here might not be applicable to other settings such as the social sciences. However, previous studies illustrate that last author positions are linked to a division of labor between senior scientists and early career researchers in the disciplines neuroblastoma research is mainly embedded in, such as medicine, epidemiology, and biomedical research (Savitz, 1999; Shapiro et al., 1994). Moreover, we did not consider shared last authorships, i.e., only the actual last author counted as the last author.

6. Methods and models

6.1. Exponential random graph models (ERGMs)

We use exponential random graph models (ERGMs) to study how neuroblastoma research changed its social structure (Lusher et al., 2013).¹³ ERGMs allow us to test whether inequality in coauthorships intensified, accumulation dynamics diversified, and segregation along the lines of status amplified throughout the field's development.

The dependent variable for ERGMs is the global structure of a given network. The independent variables are count statistics for local structures, such as the number of dyads sharing the same characteristic—e.g., researchers who collaborate and work in the same country. ERGM coefficients indicate whether a particular local structure occurs more often in the observed network than a random allocation of ties would suggest, conditional on all other local structures considered by the model specification (Lusher et al., 2013; Robins, 2011). A strength of this method is that it allows researchers to dissect the global structure of networks with a generative model, which provides parameters for local tie-formation processes while taking into account other related factors.

Another advantage of ERGMs is that they allow researchers to obtain random networks conditional on a particular model specification. Thus, global statistics capturing the structure in simulated networks can be calculated and compared with empirical values (e.g., Gondal and McLean, 2013a, 2013b). We make use of this feature to assess which models are capable of reproducing observed levels of inequality. This procedure enables us to test *Hypothesis 1*—which states that inequality in the distribution of coauthor ties increases over time—because we account for the fact that many network measures are mechanically linked to a network's opportunity structure, size, or network-endogenous tendencies (Anderson et al., 1999; Blau, 1977; Wimmer and Lewis, 2010). Also, simulations help us to evaluate whether models including researchers' characteristics provide a better approximation of observed levels of stratification, which would point to a diversification of accumulation dynamics (H2).

To implement this part of our analytical strategy, we first calculate descriptive measures capturing stratification for each conference (18 conferences from 1975 to 2016). Subsequently, we simulate 1,000 random networks—based on parameters from different ERGMs—which had the same size, density, and node attributes as the corresponding empirical network. This provides us with a distribution of statistics

stemming from simulated networks. Finally, we examine whether measures of empirical networks are substantially different from those we find in simulated networks (Gondal and McLean, 2013a, 2013b; Snijders and Steglich, 2015). While this procedure does not provide a formal test of statistical significance, it can tell us whether observed changes in network structure are substantial beyond basic network features such as changes in network density.

In addition, we compare average marginal effects (Duxbury, 2021) in different years to further investigate a potential diversification of accumulation dynamics (H2) and to probe whether segregation according to status differences became a feature of network structure as neuroblastoma research matured (H3).

6.2. Model specifications

Table 1 provides an overview of the different model specifications we estimate to study the role of researchers' characteristics for network structure over time. All specifications include the edges term, which captures the density of a network and can be thought of as intercept term reflecting the overall probability of a tie (Smith et al., 2016: 1240). Furthermore, we added terms for homophily—the tendency of similar actors to form relationships (McPherson et al., 2001)—on the country and the institutional level, because these foci are likely to shape collaborations (Dahlander and McFarland, 2013; Feld, 1981; Stark et al., 2020).

M1 is a baseline specification that only includes the terms described above and helps us to assess whether changes in network density or other basic network properties can account for changes in global inequality. The full specification (M2) adds the main effects and the absolute differences for all characteristics to account for possible interdependencies of popularity and homogeneity as well as multicollinearity between characteristics (Bojanowski and Corten, 2014; Lusher et al., 2013). Main effects reflect the popularity¹⁴ of actors according to a specific attribute, which allows us to investigate a potential diversification of accumulation dynamics (H2), because the main effects of years of experience, cumulated publications, and last author positions mirror whether researchers with a higher stock of these resources attract more collaboration partners.

To detect whether the balance between status heterogeneity and status homogeneity according to researchers' characteristics changes over time (H3), we use terms that indicate whether dissimilar dyads are more or less likely to exhibit ties. In the case of continuous attributes, the ERGM framework usually models homogeneity in reversed difference

Table 1
Summary of exponential random graph model specifications.

Model terms	M1	M2	M3	M4	M5
Edges	X	X	X	X	X
Same country	X	X	X	X	X
Same institution	X	X	X	X	X
Popularity according to experience		X	X		
Difference in authors' experience		X	X		
Popularity according to cumulated publications		X		X	
Difference in cumulated publications		X		X	
Popularity according to share last author positions		X			X
Difference in share last author positions		X			X
Reported in Table	3	4	A2	A3	A4

Note.—X signifies whether a term was included in the respective model specification.

¹³ The analysis was carried out in R. The *ergm* package was utilized to conduct the ERGM analysis (Hunter et al. 2008). In addition, the *ergMargins* package was used to calculate average marginal effects (Duxbury, 2019).

¹⁴ Please note that the network is undirected. We use the term “popularity” to denote the main effect of an attribute because collaborations are based on researchers' mutual consent, i.e., instances of declined requests to collaborate are not recorded. Technically, main effects of nodal attributes combine popularity—the tendency to receive ties—and expansiveness—the tendency to send ties—in undirected networks (Goodreau et al., 2009; Lusher et al., 2013).

scores. Therefore, a significantly negative estimate for e.g., the “Difference in authors’ experience” term reflects that scientists with similar years of experience are more likely to collaborate.

While the full specification (M2) entails all characteristics, we also estimate specifications that only add one characteristic to our baseline specification. M3 considers researchers’ years of experience, M4 entails terms for authors’ productivity, and M5 adds seniority. A comparison between M2 and M3 to M5 allows us to assess the role of researchers’ characteristics for network structure in greater depth.

6.3. Average marginal effects (AMEs)

We calculate average marginal effects (AME) as proposed by Duxbury (2019, 2021) in order to compare estimates across model specifications and conferences. This allows us to overcome methodological challenges concerning the substantial interpretation of ERGM coefficients, and the comparability of estimates between models—which can be problematic due to, e.g., residual variation (Duxbury, 2021; for similar problems in logistic regressions, see Mood, 2010). In comparison to standard coefficients, AMEs are unaffected by scaling and offer a substantial interpretation in terms of absolute changes in tie probability. For example, if the AME for the “Popularity experience” term is 0.01, this would mean that the probability of scientists with a long experience to attract an additional tie is 1 percentage point higher than for scientists with less experience.

To ensure a valid comparison of effect sizes over time, we interpret AMEs in relation to the baseline probability to form a tie during a given conference. As Kreager et al. (2021: 59, footnote 12) recently noted: “AMEs differ from odds ratios in that they are on a probability scale and so their magnitudes should be interpreted relative to the baseline tie probability (i.e., network density).”¹⁵ Therefore, we report AMEs that are divided by the baseline probability to form a tie during a given conference. These scaled AMEs can be interpreted as change of the baseline probability to form a tie if a network variable increases by one unit.

7. Results

Table 2 provides an overview of the collaboration network among neuroblastoma researchers. Whereas conferences in the first decade of the ANR series were still relatively small, with fewer than 100 authors, attendance at the conferences began to grow rapidly from the early 1990s onward. This growth was accompanied by the internationalization of the ANR series. Whereas the first eight conferences were held in Philadelphia (USA), the ANR expanded to Europe between 1998 and 2006. Then, in 2006, the conference organizers decided to rotate the ANR between the Americas, Europe, and Asia/Australia.

Parallel to the geographical expansion of the conference series, the mean degree centrality of the collaboration network increased. For instance, the early Philadelphia phase of the conference reached a maximum of—on average—13 submitted abstracts per author (1996), 20 years later this figure doubled at the ANR in Cairns in 2016. These trends are

in line with findings of a recent scientometric analysis of neuroblastoma research, which also reported the growth and proliferation of the field over time (Martynov, Klima-Frysch, and Schoenberger, 2020, see also Berthold et al., 2019).

Besides the internationalization and growth of the community, the collaboration network exhibited another pattern that provides initial evidence for the emergence of a stratified order over time (H1). The skewness of the degree distribution increased over the years, meaning that collaboration ties were distributed increasingly unequally as the field matured.

This trend is depicted in Fig. 1, which shows the network at three exemplary time points (see Table 2 for the complete history of the field). In the next section, we scrutinize the robustness of this descriptive trend and test our other theoretical considerations in the ERGM framework.

7.1. ERGM results overview

We now turn to the results of exponential random graph models. Table 3 reports estimates of the baseline specification. All tables report the average marginal effects and their corresponding delta standard errors (Duxbury, 2019, 2021).¹⁶

The full models reported in Table 4 additionally entail researchers’ characteristics—i.e., their years of experience, productivity, and seniority. Furthermore, we estimated models only adding one characteristic to the baseline specification to investigate how estimates change if characteristics are considered independently or simultaneously (reported in Tables A2, A3, and A4). We compare selected average marginal effects from these models with AMEs stemming from full models in Figs. 3 and 4.

Fig. 2 visualizes the properties of simulated networks obtained from models reported in Table 3 and Table 4. Boxplots show the distribution of simulated values, while triangles depict empirical values in each year. The upper panel of the figure allows us to assess whether inequality increased above and beyond basic network tendencies such as institutional and country homophily or changes in size and density. In addition, the bottom panel provides information on whether adding researchers’ characteristics provides better predictions for observed levels of inequality. Before we discuss our results, we provide details on the goodness of fit of our models.

7.2. Goodness of fit (GOF)

We assessed the goodness of fit (GOF) of all models by simulating networks from estimated ERGMs and comparing their degree, edgewise-shared partner, and geodesic distance statistics with the observed statistics in the corresponding network (Hunter et al., 2008). We decided to report the share of statistics with a t-ratio smaller than 2 for the degree distribution, the distribution of edgewise-shared partners, and the occurrence of geodesic distances.¹⁷ Hence, we focus on a comparison between

¹⁵ In the example sketched above, an increase by 1 percentage point can have a different substantial interpretation depending on how likely it is for a tie to form in the first place. In the case of a sparse network—exhibiting a baseline probability of e.g., 0.02—an AME of 0.01 would increase the probability to form a tie by 50%, indicating a substantial effect of experience. However, if the network’s baseline probability would be higher, e.g., 0.20, an AME of 0.01 would be less substantial, and suggest that the baseline probability to form a tie is only increased by 5%. This property of AMEs is crucial for our application since the network density of neuroblastoma researchers’ collaborations varies strongly over time.

¹⁶ As pointed out by Duxbury (2021: 8), “While rescaling does not alter conclusions about the direction and significance of noninteraction coefficients, it does affect coefficient magnitude.” In our case, the direction and significance of AMEs are identical with those of coefficients. However, due to the networks’ differing sizes, significance and direction should only be interpreted with caution.

¹⁷ Previous accounts either report GOF plots that depict the fit of all statistics of these three network properties, tables with selected statistics (e.g., Gondal and McLean, 2013), or aggregate t-ratios across network properties (e.g., Smith et al., 2016). While the first two strategies are feasible when researchers analyze a single network or a small set of networks, studies that report models for many networks must strike a balance between an in-depth report of models’ GOF and an aggregation of the fit of simulated statistics. We decided to aggregate t-ratios separately for all three network properties (Figure A1) and additionally report individual GOF plots for the degree distribution (Figure A2).

Table 2
Information on conferences and coauthor networks.

Year	City of congress	Country of congress	Size ^a	Average degree centrality	Skewness of degree distribution	Standard deviation of degree distribution
1975	Philadelphia	USA	13	0.50	1.91	0.38
1979	Philadelphia	USA	36	1.92	0.04	1.00
1984	Philadelphia	USA	97	6.00	1.76	4.20
1987	Philadelphia	USA	114	5.82	1.63	6.70
1990	Philadelphia	USA	174	5.75	1.58	4.52
1993	Philadelphia	USA	262	8.84	2.17	8.03
1994	Philadelphia	USA	205	10.13	1.25	4.12
1996	Philadelphia	USA	470	12.89	3.30	11.07
1998	Bath	UK	507	11.75	2.56	6.88
2000	Philadelphia	USA	488	12.47	2.81	8.66
2002	Paris	France	641	11.96	2.83	7.30
2004	Genoa	Italy	920	12.66	2.73	7.20
2006	Los Angeles	USA	908	14.12	3.47	8.45
2008	Chiba	Japan	970	19.51	3.12	13.84
2010	Stockholm	Sweden	1184	21.90	3.23	16.85
2012	Toronto	Canada	1117	17.00	3.32	9.93
2014	Cologne	Germany	1197	23.27	3.32	16.64
2016	Cairns	Australia	955	26.69	2.98	19.29

^a Size refers to the number of authors who participated at least twice in the ANR conference series.

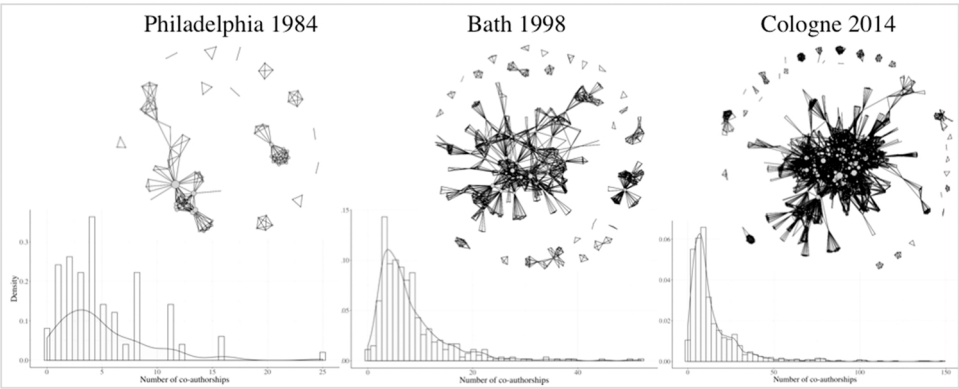


Fig. 1. The changing structure of the coauthor network at three time points. Note.—Nodes in the sociograms depict researchers who participated at least twice in the ANR conference series; ties between them indicate that they appeared as coauthors on an abstract submitted to the ANR conference. Histograms are based on the distribution of coauthor ties in the relevant year. The x-axis denotes the number of coauthor ties per author and the y-axis the density of the distribution.

our baseline model specification and full models. In addition, we report plots depicting the GOF for networks’ degree distributions. This allows us to explore which model specification better accounts for actors holding many collaborations. GOF analyses are reported and discussed in Appendix A.

In summary, results indicate that the GOF for edgewise-shared partners and geodesic distances was insufficient, irrespective of the model specification. In contrast, the fit for the degree distribution was noticeably better in full models, especially in the later stages of the

field’s development. These findings corroborate our theoretical expectation that the importance of researchers’ characteristics for the distribution of collaboration ties increases over time.

The remainder of the results section will proceed as follows. First, we discuss the results of simulations to map changing stratification in coauthorships. Second, we turn to the questions of whether accumulation dynamics diversified and status homogeneity increased over time.

Table 3

Average marginal effects (AMEs) of baseline exponential random graph models (ERGMs) for collaboration network.

Years	1979		1984		1987		1990		1993		1994	
	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	-0.004 (0.021)	-0.094	0.094*** (0.008)	1.793	0.061*** (0.006)	0.996	0.075*** (0.003)	2.273	0.049*** (0.002)	1.594	0.080*** (0.003)	2.441
Same institution	0.110*** (0.017)	2.754	0.091*** (0.008)	1.739	0.095*** (0.009)	1.548	0.053*** (0.003)	1.597	0.055*** (0.003)	1.768	0.030*** (0.003)	0.931
Baseline probability	0.040		0.053		0.062		0.033		0.031		0.032	
Years	1996		1998		2000		2002		2004		2006	
	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	0.039*** (0.001)	1.895	0.034*** (0.001)	2.121	0.030*** (0.001)	1.677	0.031*** (0.001)	2.276	0.022*** (0.0004)	2.347	0.020*** (0.0004)	1.911
Same institution	0.034*** (0.001)	1.656	0.027*** (0.001)	1.721	0.035*** (0.001)	1.971	0.025*** (0.001)	1.877	0.020*** (0.0003)	2.107	0.024*** (0.0003)	2.291
Baseline probability	0.021		0.016		0.018		0.013		0.009		0.010	
Years	2008		2010		2012		2014		2016			
	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	0.021*** (0.0004)	1.704	0.021*** (0.0003)	1.775	0.018*** (0.0003)	2.012	0.023*** (0.0003)	1.881	0.029*** (0.0004)			1.630
Same institution	0.026*** (0.0004)	2.052	0.025*** (0.0003)	2.089	0.021*** (0.0003)	2.291	0.026*** (0.0003)	2.089	0.036*** (0.0005)			1.980
Baseline probability	0.013		0.011		0.009		0.012		0.018			

Note.—All continuous variables are z-standardized to enhance the comparability of estimates across models. Delta standard errors (Duxbury, 2019) are reported in parentheses. Scaled AMEs are AMEs divided by the baseline probability and can be interpreted as relative changes in tie probability if a network variable increases by one unit. We multiplied scaled AMEs by 100 to provide a measure capturing the percentage change of the baseline probability in Fig. 3 and Fig. 4.

† p < 0.10 * p < 0.05 ** p < 0.01 *** p < 0.001 (two-sided)

Table 4
Average marginal effects (AMEs) of full exponential random graph models (ERGMs) for collaboration network.

Years	1979		1984		1987		1990		1993		1994	
	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	-0.004 (0.021)	-0.111	0.097*** (0.008)	1.836	0.065*** (0.006)	1.062	0.074*** (0.003)	2.258	0.045*** (0.002)	1.463	0.08*** (0.003)	2.353
Same institution	0.105*** (0.017)	2.643	0.091*** (0.008)	1.720	0.090*** (0.009)	1.470	0.054*** (0.003)	1.648	0.060*** (0.003)	1.938	0.03*** (0.003)	1.038
Popularity years of experience	0.001 (0.006)	0.034	-0.009** (0.003)	-0.177	0.006* (0.003)	0.091	0.002 (0.001)	0.065	0.005*** (0.001)	0.159	0.002 (0.001)	0.057
Difference in authors' years of experience	-0.007 (0.009)	-0.174	-9e-03 (0.004)	-0.018	-0.007* (0.003)	-0.118	-0.005** (0.002)	-0.140	-0.004*** (0.001)	-0.138	-0.005** (0.002)	-0.147
Popularity cumulated publications	0.008 (0.011)	0.206	0.007 (0.006)	0.128	0.014*** (0.003)	0.227	0.007*** (0.001)	0.198	0.009*** (0.001)	0.278	0.005** (0.001)	0.168
Difference in cumulated publications	-0.024 (0.017)	-0.595	0.005 (0.007)	0.085	0.006 (0.003)	0.093	0.002 (0.002)	0.064	-0.002* (0.001)	-0.079	-0.001 (0.002)	-0.038
Popularity share last author positions	0.001 (0.007)	0.028	-0.008 (0.006)	-0.160	-0.013** (0.005)	-0.210	-0.015*** (0.003)	-0.466	-0.005*** (0.001)	-0.170	-0.01*** (0.002)	-0.243
Difference in share last author positions	0.003 (0.009)	0.067	0.008 (0.006)	0.190	0.004 (0.006)	0.072	0.013*** (0.003)	0.389	0.001 (0.002)	0.039	0.007** (0.002)	0.200
Baseline probability	0.040		0.053		0.062		0.033		0.031		0.032	
Years	1996		1998		2000		2002		2004		2006	
	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	0.037*** (0.001)	1.802	0.032*** (0.001)	2.035	0.029*** (0.001)	1.616	0.029*** (0.001)	2.184	0.021*** (0.0004)	2.280	0.019*** (0.0004)	1.853
Same institution	0.036*** (0.001)	1.744	0.028*** (0.001)	1.770	0.036*** (0.001)	1.990	0.025*** (0.001)	1.897	0.020*** (0.0003)	2.090	0.023*** (0.0003)	2.262
Popularity years of experience	0.0001 (0.0003)	0.006	0.0007* (0.0003)	0.044	0.002*** (0.0003)	0.126	0.002*** (0.0002)	0.139	0.001*** (0.0001)	0.086	0.001*** (0.0001)	0.120
Difference in authors' years of experience	-0.004*** (0.001)	-0.184	-0.005*** (0.0004)	-0.340	-0.004*** (0.0004)	-0.243	-0.003*** (0.0003)	-0.213	-0.002*** (0.0002)	-0.164	-0.002*** (0.0002)	-0.178
Popularity cumulated publications	0.010*** (0.0003)	0.462	0.006*** (0.0003)	0.353	0.006*** (0.0004)	0.334	0.003*** (0.0003)	0.216	0.002*** (0.0002)	0.195	0.003*** (0.0002)	0.305
Difference in cumulated publications	-0.002*** (0.0004)	-0.082	>0.0001 (0.0004)	0.016	-0.002*** (0.0004)	-0.081	-0.0004 (0.0003)	-0.020	0.0002 (0.0002)	0.017	-0.001*** (0.0002)	-0.069
Popularity share last author positions	-0.003*** (0.001)	-0.145	-0.002*** (0.001)	-0.151	-0.003*** (0.001)	-0.190	-0.0003 (0.0003)	-0.024	-0.001*** (0.0002)	-0.085	-0.001*** (0.0002)	-0.121
Difference in share last author positions	>0.001 (0.001)	0.026	0.002*** (0.001)	0.128	0.003*** (0.001)	0.160	0.001* (0.0004)	0.064	0.001** (0.0002)	0.068	0.001* (0.0003)	0.066
Baseline probability	0.021		0.016		0.018		0.013		0.009		0.010	

(continued on next page)

Table 4 (continued)

Years	2008		2010		2012		2014		2016	
	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	0.020*** (0.0004)	1.620	0.019*** (0.0003)	1.633	0.017***	1.894	0.022*** (0.0003)	1.794	0.028*** (0.0004)	1.571
Same institution	0.026*** (0.0004)	2.072	0.025*** (0.0003)	2.163	0.021***	2.295	0.026*** (0.0003)	2.115	0.037*** (0.001)	2.020
Popularity years of experience	0.001*** (0.0001)	0.109	0.001*** (0.0001)	0.120	0.0004***	0.046	0.001*** (0.0001)	0.099	0.003*** (0.0002)	0.145
Difference in authors' years of experience	-0.002*** (0.0002)	-0.178	-0.002*** (0.0002)	-0.187	-0.001***	-0.114	-0.002*** (0.0002)	-0.129	-0.003*** (0.0002)	-0.142
Popularity cumulated publications	0.004*** (0.0002)	0.345	0.005*** (0.0001)	0.380	0.003***	0.345	0.005*** (0.0001)	0.365	0.006*** (0.0002)	0.343
Difference in cumulated publications	-0.001*** (0.0002)	-0.072	-0.002*** (0.0001)	-0.131	-0.001***	-0.084	-0.001*** (0.0001)	-0.104	-0.002*** (0.0002)	-0.110
Popularity share last author positions	-0.001*** (0.0002)	-0.080	-0.0004* (0.0002)	-0.045	-0.0004*	-0.060	-0.0004* (0.0002)	-0.031	-0.002*** (0.0003)	-0.102
Difference in share last author positions	-0.0004 (0.0003)	-0.015	-0.0005* (0.0002)	-0.042	0.0003	0.035	-0.0005* (0.0002)	-0.043	-0.001 (0.0004)	-0.038
Baseline probability	0.013		0.011		0.009		0.012		0.018	

Note.—All continuous variables are z-standardized to enhance the comparability of estimates across models. Delta standard errors (Duxbury, 2019) are reported in parentheses. Scaled AMEs are AMEs divided by the baseline probability and can be interpreted as relative changes in tie probability if a network variable increases by one unit. We multiplied scaled AMEs by 100 to provide a measure capturing the percentage change of the baseline probability in Fig. 3 and Fig. 4. † $p < 0.10$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$ (two-sided)

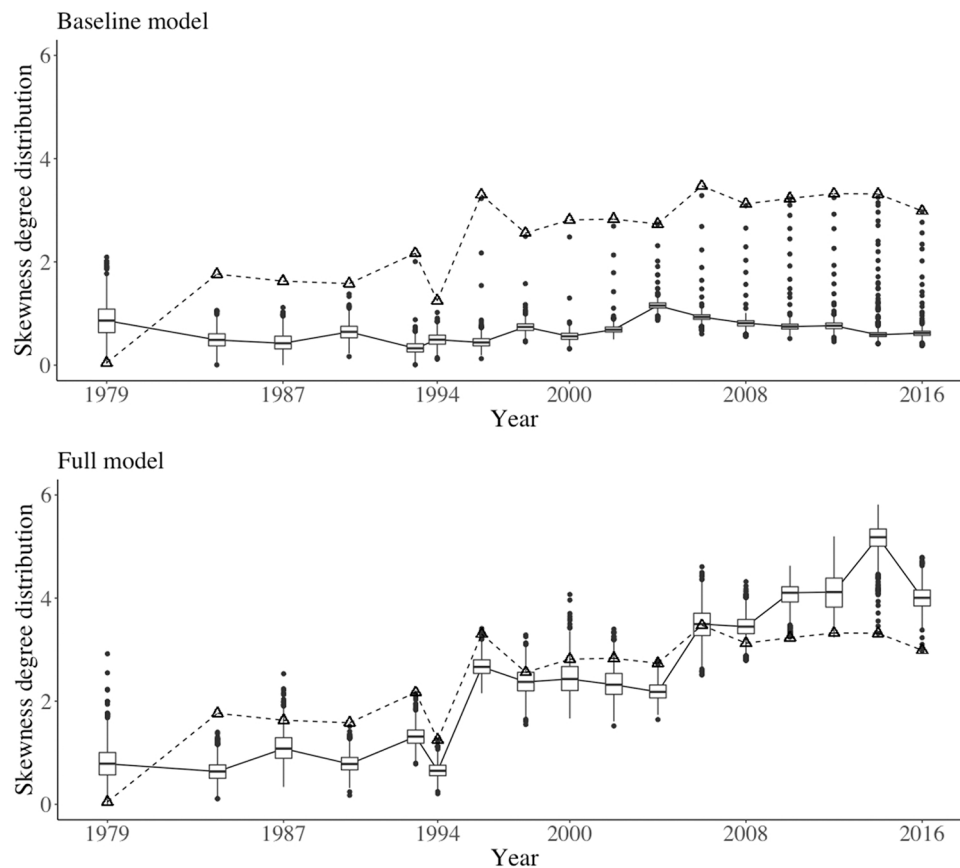


Fig. 2. Predicting inequality in coauthor ties over time Note.—Each box plot represents the distribution of skewness values calculated in simulated networks obtained from network models reported in Tables 3 and 4. Boxplots in the upper panel show simulated values for networks generated according to our baseline model specification (see Table 3). The bottom panel reports simulated values generated from the full models that include researchers' characteristics (Table 4). We simulated 1,000 networks for each year and specification. Triangles indicate empirical values, i.e., the observed skewness of the degree distribution in a particular year. The dashed line connects empirical values, while the straight line follows the medians of simulated values.

7.3. Rising inequality in coauthor ties

We simulated 1,000 networks for each conference from baseline models. Simulated networks had the same number of nodes and the same densities as observed networks. Thus, they tell us whether an unequal degree distribution could also have been a by-product of basic network properties such as changes in the average size of research teams. Fig. 2 summarizes the skewness in simulated networks in box plots and shows how empirical values—depicted by triangles—differed from simulated values.¹⁸ While simulated networks stemming from baseline models indicated a mild trend toward more inequality, empirical values differed starkly from 1996 onward and experienced a jump in inequality after the first 20 years of the community's history. Another, smaller, increase in inequality was visible after 2004. Overall, these results are in line with our first theoretical expectation that inequality in the distribution of coauthor ties increases as a scientific field matures and grows (H1).

Furthermore, a comparison between the upper and the bottom panel provides further evidence for our second hypothesis. While simulations obtained from baseline models were not sufficient to approximate the empirical trend in inequality, full models substantially improved our

predictions, especially in the later stages of the field's development. A similar tendency is visible if we compare the fit of degree distributions between baseline and full models over time (see Fig. A2 in Appendix A). In the next section, we provide a more detailed picture of how researchers' characteristics mold network structure over time.

7.4. Diversification of accumulation dynamics

Here, we consider whether a scientist's prominence as a coauthor was linked to a variety of characteristics as the field matured (H2).

Fig. 3 depicts selected average marginal effects (AMEs) and their corresponding confidence intervals stemming from full models (Table 4) and from models only adding one characteristic to the baseline specification (reported in Table A2 for years of experience, Table A3 for productivity, and Table A4 for seniority). AMEs from full models are represented by dots, while diamonds visualize AMEs obtained from simpler models.

The results of specifications that add only one characteristic to the baseline specification indicate that researchers with one standard deviation more years of experience and last author positions exhibited significantly more coauthorships than their peers in the second half of the field's development. For instance, in 1994, researchers with one standard deviation more years of experience showed a 15.5% higher probability of engaging in an additional collaboration relative to the baseline probability of forming a tie during this year.

¹⁸ We repeated this analysis with the standard deviation (e.g., Snijders and Steglich 2015) and the Gini coefficient (Badham, 2013) as additional measures for inequality. The results for different measures were qualitatively similar and are reported in Appendix A (Figures A3 and A4).

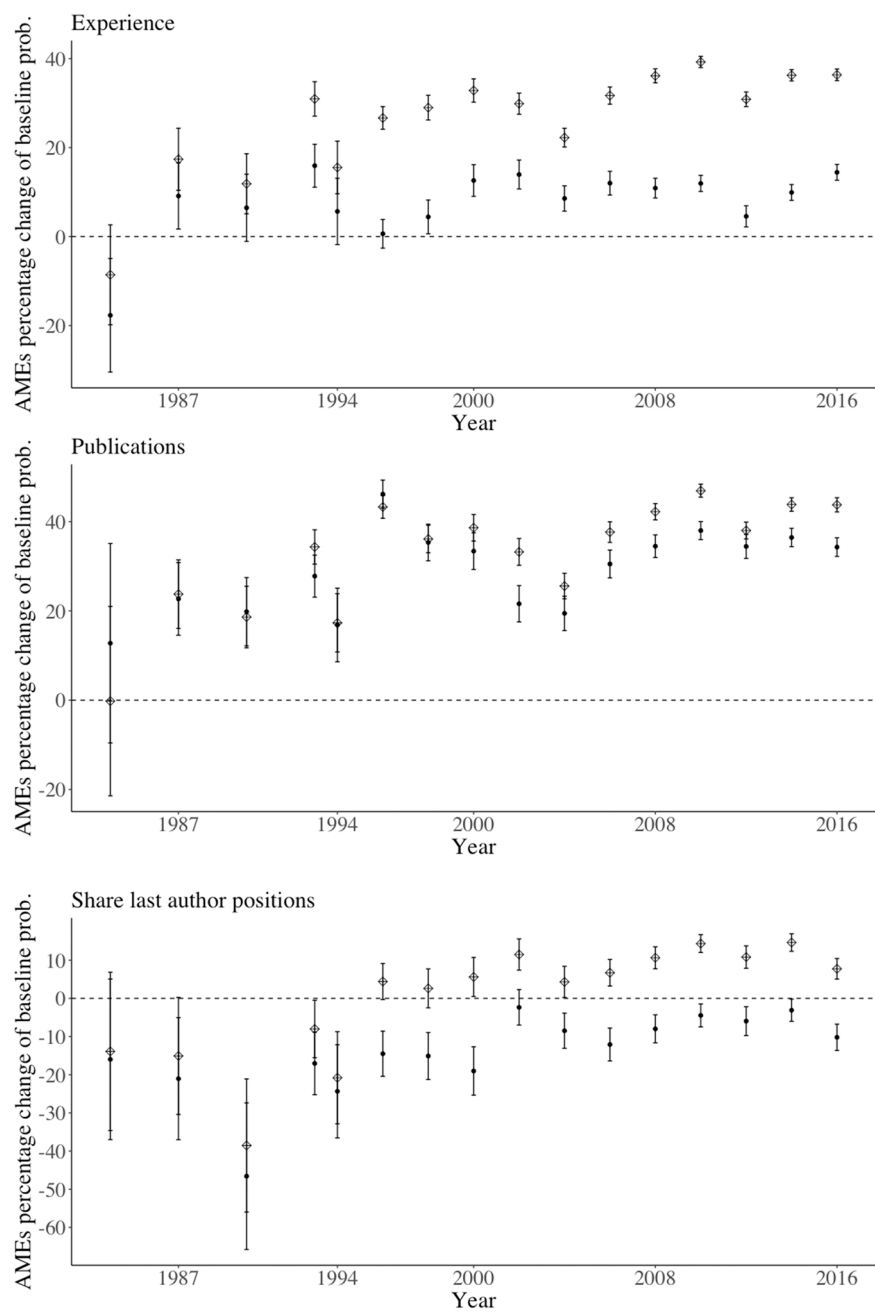


Fig. 3. Popularity according to researchers’ characteristics over time. Note.—Dots represent scaled AMEs and their 95% confidence intervals for main effects derived from full models in Table 4. Diamonds depict coefficients from model specifications that only include one of the researchers’ characteristics in addition to the baseline specification. These models are reported in Appendix A (Tables A2, A3, and A4).

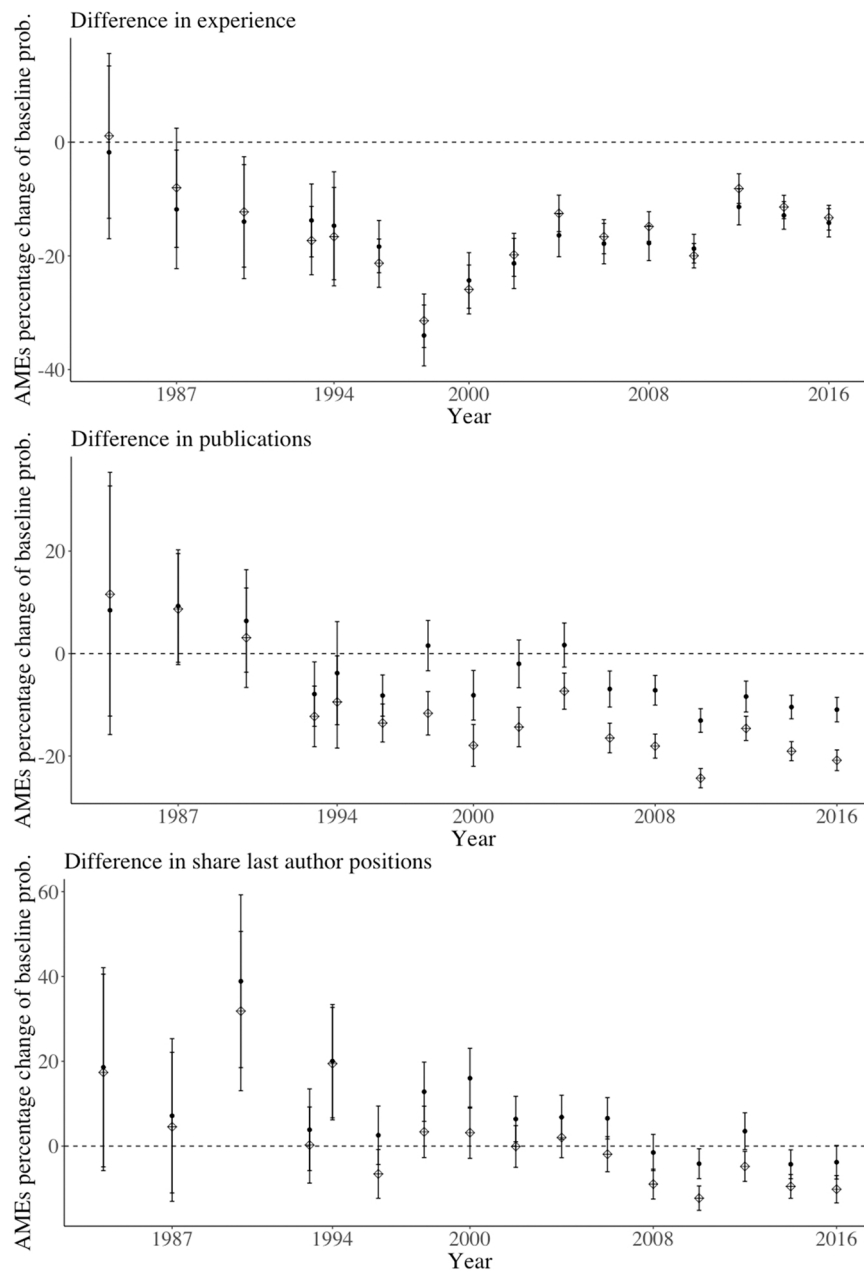


Fig. 4. Homogeneity according to researchers' characteristics over time. Note.—Dots represent scaled AMEs and their 95% confidence intervals for z-standardized difference scores derived from our full models in Table 4. Diamonds depict AMEs from model specifications that only include one of the researchers' characteristics in addition to the baseline specification. These models are reported fully in Appendix A (Tables A2, A3, and A4).

From 1996 to 2006, one standard deviation more experience elevated the baseline probability by more than 22%, and, from 2008 to 2016, by at least 31%.¹⁹

Similarly, productivity and seniority showed a rising association with more coauthor ties from 1996 onward. Yet, for seniority, AMEs obtained from full models show that including multiple characteristics simultaneously changes this picture: the AMEs for last author positions

showed negative values in all years in M2 (Table 4). This pattern points to fewer coauthorships held by senior researchers after accounting for their experience and productivity.

7.5. Status homogeneity over time

In this section, we test our third theoretical expectation that new scientific fields are mainly characterized by status-dissimilar collaborations, whereas later stages of a field should additionally display circles of co-authors who are status-similar (H3). We included model terms that capture whether a one standard deviation difference in years of experience, productivity, or seniority changes two scientists' likelihood of collaborating.

Fig. 4 shows how the tendency to affiliate with others of similar status in terms of years of experience, publications, and share of last author positions changed over time. Consistent with the notion that a field is less marked by closure along differences in status at its outset,

¹⁹ These are AMEs for the specification which only included years of experience as researcher characteristic (M3, reported in Table A2 and depicted as diamonds in Fig. 3). The corresponding numbers from full models are as follows: 1994: 6%; 1996–2006: at least 1%; 2008–2016: at least 5% (M2, reported in Table 4 and depicted as dots in Fig. 3). While these numbers indicate a decreased importance of experience once other characteristics are considered, a time trend towards rising AMEs in the second half of the field's development is still visible in Fig. 3.

average marginal effects showed a decreasing trend.

Here, decreasing AMEs mean that collaborative ties between status-dissimilar partners became less likely as the field matured. For instance, a one standard deviation difference in publications showed positive AMEs from 1984 to 1990. Subsequently, the results indicate that a one standard deviation difference in publications reduced scientists' baseline probability to collaborate by 7% to 18% between 1993 and 2004. In the third stage of the field's development (2006–2016), baseline probabilities to collaborate were reduced by 15% to 24% for researchers with dissimilar publication records.²⁰

In a similar vein, differences in years of experience decreased scientists' probability to collaborate but started to structure collaboration networks earlier than scientists' publication records. Also, a curvilinear trend is visible: AMEs decreased until 1998—indicating elevated importance of differences in experience for tie-formation—and increased afterward.

For the simple model specification (M5), AMEs for differences in seniority showed a similar trend as differences in publications indicating that a tie between dissimilar researchers became less likely over time. In addition, AMEs for seniority are relatively small and only show negative values during a late stage of the field's development. Between 2008 and 2016, scaled AMEs suggest a reduction of the baseline probability between 5% and 10% (M5, reported in Table A4). The corresponding values for the full model specification range from 1.5% to 4% (M2, reported in Table 4).

In summary, our results suggest that status-similar collaborations became more likely over time. However, collaborations between early career researchers and established scientists probably retained their relevance for scientific work in neuroblastoma research. The analyses presented here did not investigate whether high-degree nodes are connected to low-degree nodes (assortativity). Instead, we provided estimates for status homogeneity net of other model terms and the opportunity structure to form ties. Moreover, the large AMEs for institutional homophily could partially point to the persisting importance of mentor-apprentice ties parallel to the trends toward status homogeneity according to researchers' attributes. We will further discuss this aspect in the concluding section of the article.

7.6. Additional analyses

In addition to ERGMs, we also estimated stochastic actor-oriented models (SAOMs, Snijders et al., 2010), which we report in an online supplement. Due to the rapidly changing size of the network, a violation of the assumption that actors have a sense of all potential collaboration partners (Ripley et al., 2019), and issues with model fit and convergence, we were unable to make comparisons across the entire history of the field using SAOMs. Nevertheless, the results from longitudinal network models show overall consistency with the results reported in the main text, which further strengthens our confidence that the new patterns we detected with our analyses are not a mere by-product of the network's changing size and density or other basic network tendencies such as homophily according to social foci (Anderson et al., 1999; Feld, 1981).

8. Discussion

How do stratified orders in new scientific fields evolve? To address this question, we analyzed a unique data set spanning 41 years of scientific collaboration in neuroblastoma research. We integrated previous accounts that assume a link between the importance of status and the amount of uncertainty exhibited by social environments (Lynn, 2014; Mayhew, 1973; McFarland et al., 2014; Podolny, 2010; Ridgeway, 2019; Ridgeway and Erickson, 2000) with research highlighting the role of

various resources in mobilizing collaboration partners (Chubin, 1976; Frickel and Gross, 2005; Griffith and Mullins, 1972; Knorr, 1999; Latour and Woolgar, 1986; Lazega et al., 2016, 2006; Li et al., 2013). Thus, we derived hypotheses about the developmental trajectory of a growing field. We expected increasing inequality in the distribution of coauthorships, a diversification of accumulation dynamics, and rising segregation along status differences. Our results supported our hypotheses.

We found increasing inequality in the distribution of coauthorships, suggesting the formation of an elite of authors at the center of the network. Simulations obtained from network models enabled us to confirm that this trend is substantial (Gondal and McLean, 2013a; Snijders and Steglich, 2015). Regarding our expectation that circles of status-similar authors accompany a mentor-apprentice model of collaboration over time, we found a trend toward more status homogeneity according to researchers' experience and productivity. Yet, we would like to point out that our ERGM analysis focused on researchers' characteristics and did not investigate whether actors with many ties collaborate with actors holding few ties (assortativity). In comparison, results of SAOMs suggest that, independent of their attributes, researchers with many coauthor ties tended to collaborate with others who hold fewer ties in most periods. This finding supports the idea that a mentor-apprentice model of collaboration was simultaneously present to status homogeneity according to researchers' attributes (for details, see online Appendix B). Furthermore, the large estimates for institutional homophily in ERGMs and SAOMs can be interpreted as suggestive evidence for a persisting relevance of local interaction among senior scientists and early career researchers.

In spite of the insights we have provided, we also acknowledge several limitations that should be addressed in further research. While our data set is a comprehensive documentation of collaboration in neuroblastoma research, it lacks fine-grained information regarding individual researchers. For instance, we were unable to establish how researchers' gender affects collaboration (Bozeman and Gaughan, 2011; Holman and Morandin, 2019; Main, 2014) and could not account for the funding that individuals managed to accumulate (Bol et al., 2018). Likewise, we had no information about scientists' activities before they engaged with neuroblastoma research, such as their educational careers or their disciplinary background. Therefore, we focused on the link between field-specific resources and the structure of collaborations because previous accounts suggest that scientists can seldom import resources from adjacent fields into new fields of inquiry (Bourdieu, 1988; Griffith and Mullins, 1972; Latour, 1987). Additional information would have allowed us to study how the allocation of coauthorships differed over time—depending on who entered the field—instead of the linear trend towards more diversification that we tested here. Future research could expand our efforts by studying non-linear trajectories of the diversification of accumulation dynamics in scientific communities.

Similarly, further research could investigate which network mechanisms are dominant in producing the observed trend toward more inequality in the distribution of coauthor ties. Our analyses established that the trends towards more stratification and segregation are substantial and are connected to researchers' accumulated status in terms of experience, productivity, and seniority. Yet, further studies could utilize future advancements in longitudinal network modelling to deepen our understanding of how network mechanisms interact with one another to produce different global outcomes.²¹

Another limitation was that we had no information on the content of

²⁰ These are AMEs for the specification that only added accumulated publications to the baseline specification (M4, reported in Table A3 and depicted as diamonds in Fig. 4). The corresponding numbers from the full models are as follows: 1993–2004: 1.6%–8%; 2006–2016: 7%–12% (M2, reported in Table 4 and depicted as dots in Fig. 4).

²¹ Our initial analytical approach was to apply SAOMs and to simulate how global levels of inequality and segregation change if parameters for researchers' characteristics are manipulated (cf., Adams and Schaefer, 2016; Snijders and Steglich, 2015). The rapid growth of the network, a violation of the assumption that actors consider all potential collaboration partners (Ripley et al., 2019), and issues with model fit and convergence, forced us to abandon this strategy. We hope that new developments in modeling large networks with SAOMs will solve these issues (for details, see online Appendix B).

the research conducted by the scientists in our study. Therefore, we did not investigate how changes in the social structure of neuroblastoma research were accompanied by shifts in knowledge production (Chubin, 1976; Cole and Harriet, 1975; Griffith and Mullins, 1972; Knorr, 1999; Latour and Woolgar, 1986; Mullins et al., 1977). However, it is noteworthy that the structural changes towards more stratification and segregation according to status differences that we observed took place roughly in the second half of the field's history. This later stage in the field's history was also characterized by an influx of researchers with a background in molecular biology (Brodeur, 2003; Martynov et al., 2020) and a steady internationalization of the community (Berthold et al., 2019). These new members widened neuroblastoma research's interdisciplinary scope and promoted the field in more countries. Simultaneously, new insights were produced from the late 1990s onward, such as a deeper understanding of the disease's genetic mechanisms (Brodeur, 2003; Mossé et al., 2008; Ray, 2019) and improved treatment strategies (Maris, 2010; Matthay et al., 1999). Whether advancements in scientific knowledge are linked to changes in stratification and segregation along status differences in coauthor networks are exciting questions for further research.

A further problem was that we could not consider the role of institutions to its full extent. As previous research shows, academic institutions often have a distinct status of their own and this affects the success and collaborative choices of the individuals affiliated with them (Burris, 2004; Gondal, 2018; Hagstrom, 1971; Han, 2003; Lazega et al., 2006; Ma et al., 2020). Moreover, large institutes tend to offer researchers more resources to conduct their research and play a key role in understanding the formation of collaboration among scientists (Latour and Woolgar, 1986; Lazega et al., 2006).

Furthermore, as we focused on one particular scientific field, we cannot separate general developments in science from trends that are specific to the community of neuroblastoma researchers. For instance, the reported diversification of accumulation dynamics could be due to an overall trend toward the economization and increasing stratification

of knowledge production over the last few decades (Evans, 2008; Fochler et al., 2016; Jones et al., 2008; Leahey and Barringer, 2020; Lok, 2016; Münch, 2014; Münch and Baier, 2012). Further research should therefore compare the trajectories of different fields to help us understand what changes typically occur as fields mature and what developments are common across fields, for example, due to macroeconomic trends (Ramage et al., 2020; Stark et al., 2020).

Despite these limitations, our study contributes to a better understanding of how stratified orders emerge in new scientific fields. Our theoretical considerations provide a relational and dynamic view of inequality and segregation in emerging scientific fields. Moreover, we propose a complementary perspective to studying individual trajectories (e.g., Azoulay et al., 2010; Costas and Bordons, 2011; Hăncean et al., 2021; Li et al., 2013; Petersen et al., 2014), macrorends within and across scientific disciplines (Evans, 2008; Foster et al., 2015; Jones et al., 2008; Leahey, 2016; Moody, 2004; Münch, 2014; Shi et al., 2015; Wuchty et al., 2007), or the micro-interactional antecedents for knowledge production (Latour and Woolgar, 1986; Parker and Hackett, 2012) by focusing on long-term changes in the overall structure of scientific collaboration in demarcated fields of research. Finally, we have shown how our theoretical considerations apply to empirical settings by mapping how the structure of collaboration changed among neuroblastoma researchers.

Acknowledgements

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Appendix A

Table A1

Table A1
Information on scientists' characteristics.

Year	Years of experience					Number of publications per year					Seniority—share of last author positions				
	Mean	SD	Median	Min.	Max.	Mean	SD	Median	Min.	Max.	Mean	SD	Median	Min.	Max.
1975	0	0	0	0	0	0.69	1.65	0	0	6	0.15	0.38	0	0	1
1979	1.22	1.89	0	0	4	1.78	3.10	0.50	0	16	0.27	0.39	0	0	1
1984	1.73	2.89	0	0	9	0.45	1.20	0	0	7	0.17	0.35	0	0	1
1987	2.86	3.56	3	0	12	2.17	2.86	2.17	0	12	0.14	0.32	0	0	1
1990	2.02	3.63	0	0	15	1.96	2.67	1	0	11	0.19	0.36	0	0	1
1993	3.00	4.08	3	0	18	4.27	5.13	3	0	24	0.17	0.35	0	0	1
1994	2.23	3.34	1	0	19	3.76	5.63	2	0	28	0.16	0.33	0	0	1
1996	3.49	4.42	2	0	21	5.10	6.81	3	0	40	0.13	0.30	0	0	1
1998	3.66	4.51	3.66	0	23	6.18	8.58	3	0	63	0.14	0.30	0	0	1
2000	4.30	4.81	2	0	25	7.96	11.33	4	0	88	0.12	0.28	0	0	1
2002	4.30	5.10	4.30	0	27	7.56	12.62	3	0	110	0.12	0.27	0	0	1
2004	4.14	5.26	2	0	29	6.99	12.91	2	0	113	0.12	0.28	0	0	1
2006	4.69	5.53	2	0	31	7.37	13.83	3	0	125	0.13	0.30	0	0	1
2008	5.53	5.96	4	0	29	8.42	15.54	3	0	137	0.11	0.27	0	0	1
2010	5.51	6.33	4	0	31	8.35	16.46	3	0	155	0.10	0.25	0	0	1
2012	5.86	6.51	4	0	33	9.20	18.71	2	0	170	0.12	0.28	0	0	1
2014	6.64	6.86	4	0	35	9.67	19.95	3	0	183	0.09	0.25	0	0	1
2016	8.99	7.18	6	0	37	13.17	23.96	5	0	201	0.10	0.25	0	0	1

Goodness of fit (GOF): general trends

Fig. A1 reports the share of simulated statistics for the distribution of edgewise-shared partners, geodesic distances, and degrees that showed a tolerable fit in relation to empirical statistics. Nearly all models showed an inappropriate GOF for geodesic distances and edgewise-shared partner statistics. Also, the fit for these network properties did not improve after adding researchers' characteristics to the models. In contrast, the fit for the degree distribution was improved by accounting for researchers' characteristics, especially during the second half of the field's development. This result is in line with our theoretical

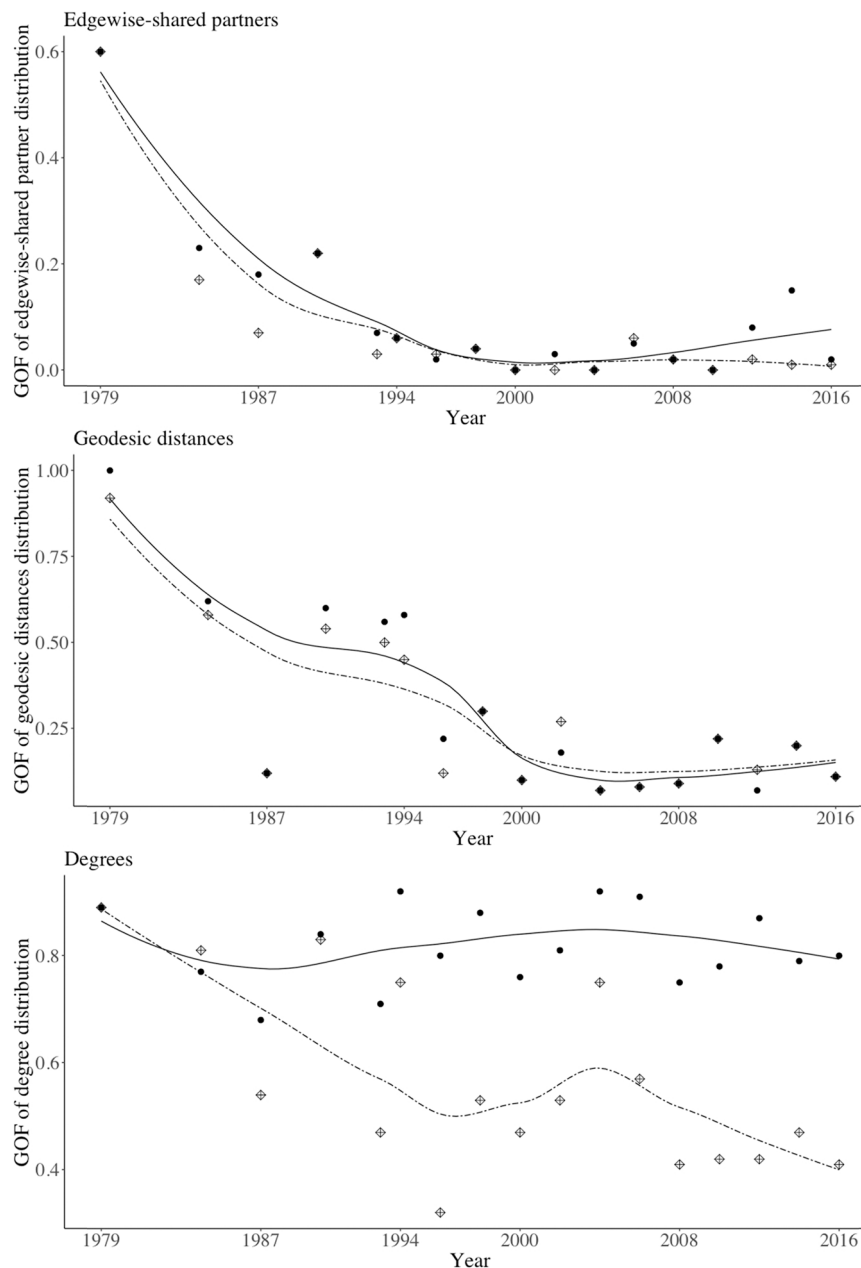


Fig. A1. Difference in goodness of fit (GOF) between baseline and full model specification. Note.—Each diamond and dot displays the share of statistics with a t -value below 2. A higher share indicates better model fit. Diamonds represent the GOF calculated for baseline models reported in Table 3. Dots depict the GOF derived for full models reported in Table 4. Lines are loess curves to enhance interpretability (dotted lines for baseline models, straight lines for full models).

expectation that researchers' characteristics became more important for collaboration as the field matured (H2).

In addition, we would like to point out that an insufficient GOF for the distribution of edgewise-shared partners and geodesic distances is not unusual in large networks (similar issues are reported for SAOMs by Lewis and Kaufman, 2018: 1736, Stark et al., 2020: 458). We attempted to increase the GOF by adding geometrically weighted statistics—such as the GWDEG and GWESP terms (Hunter, 2007). Yet, these terms led to model degeneracy in several years.²² Consequently, we decided to

report specifications that worked for all conferences instead. While a high GOF is desirable, simpler specifications sufficiently addressed our research questions regarding the link between authors' attributes and the distribution of coauthorships. Moreover, terms beyond dyadic configurations introduce complex interdependencies among parameters and can complicate interpretation (Martin, 2020; Rubineau et al., 2019).

Goodness of fit (GOF): detailed discussion of fitting the degree distribution

Fig. A2 further explores differences in the goodness of fit between baseline and full models for networks' degree distributions. Three major trends can be inferred from the figure

First, the difference between the simulated degree distributions produced by baseline models and full models becomes more pronounced over time. This is in line with the bottom panel of Fig. A1, which

²² While models that operate only on the dyad level use pseudo-maximum likelihood estimation, models that include terms beyond dyadic interdependence rely on Monte Carlo Markov Chains (Hunter et al., 2008). The latter simulation-based estimation procedure probably caused model instability in the networks under study. Also, please note that we do not report information on convergence t -ratios because these are only calculated if ERGMs are estimated by a simulation-based procedure.

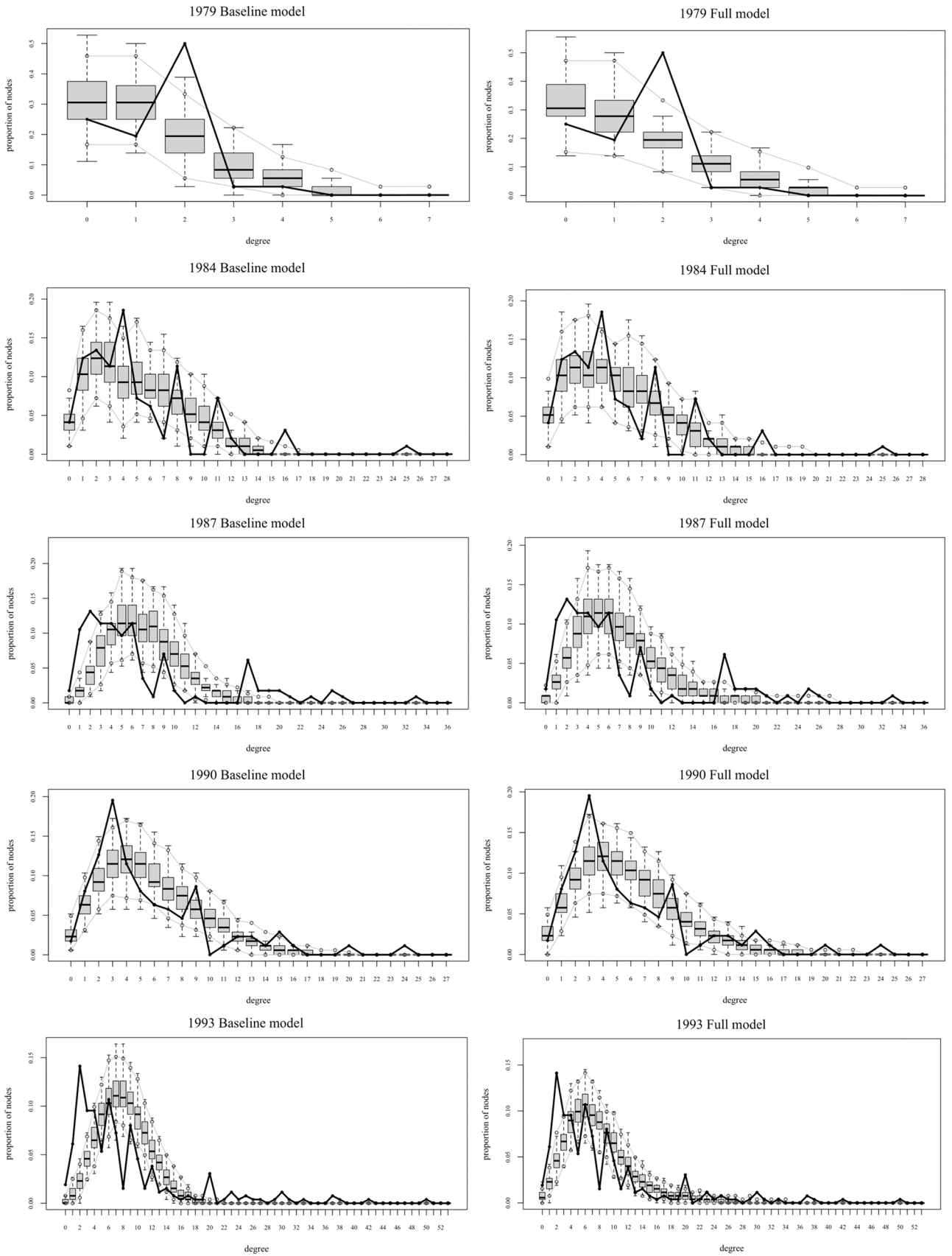


Fig. A2. Goodness of fit (GOF) plots for the degree distribution.

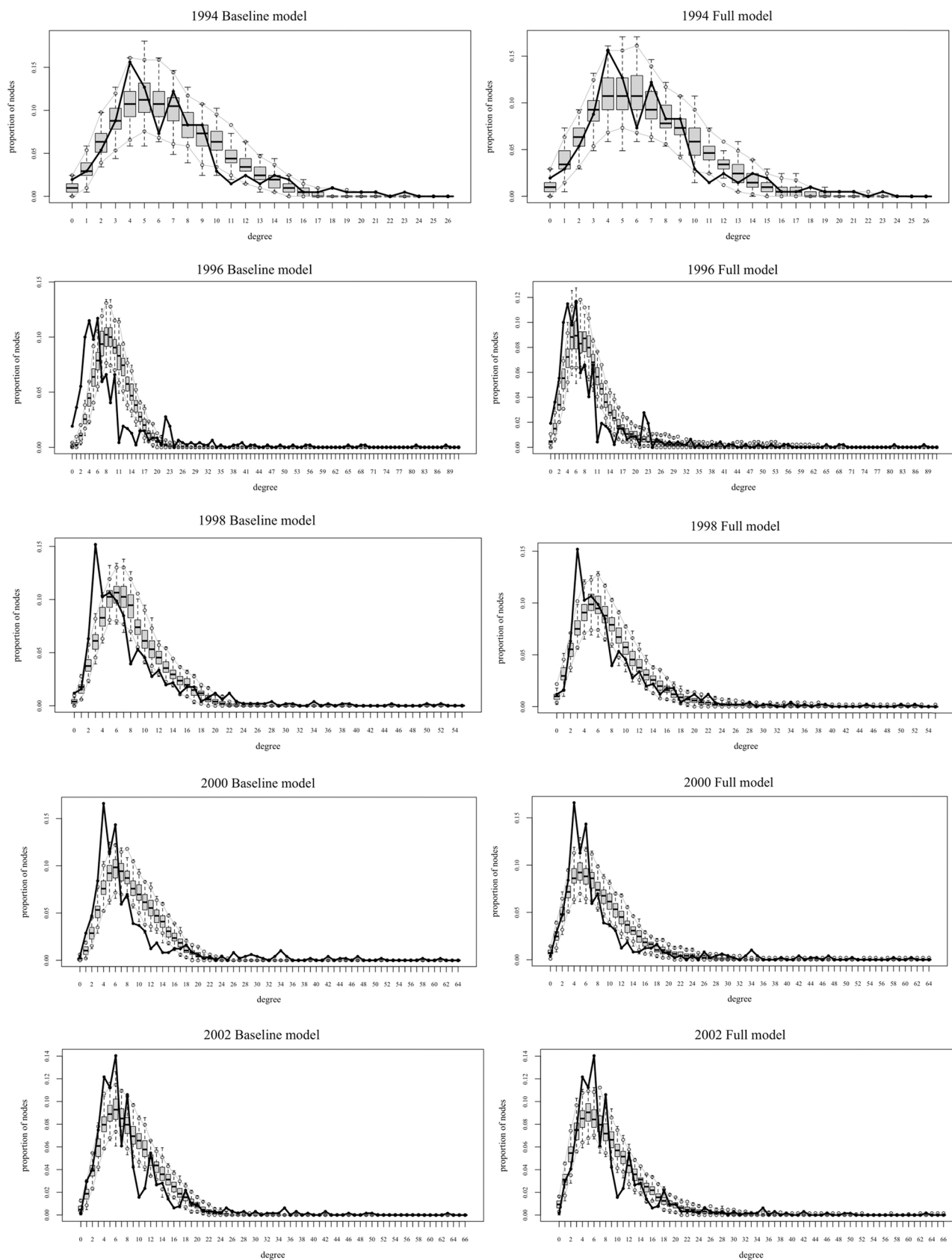


Fig. A2. (continued).

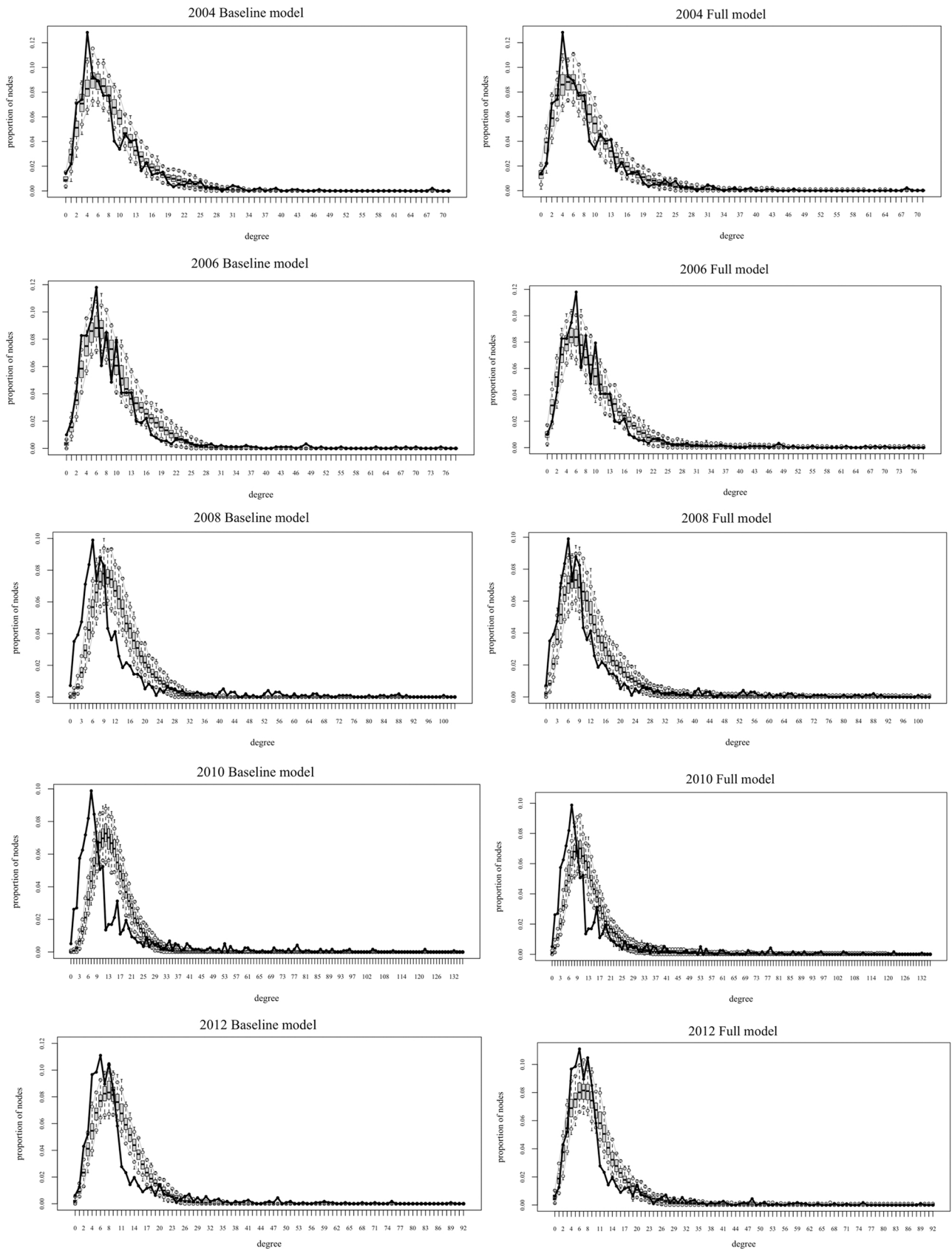


Fig. A2. (continued).

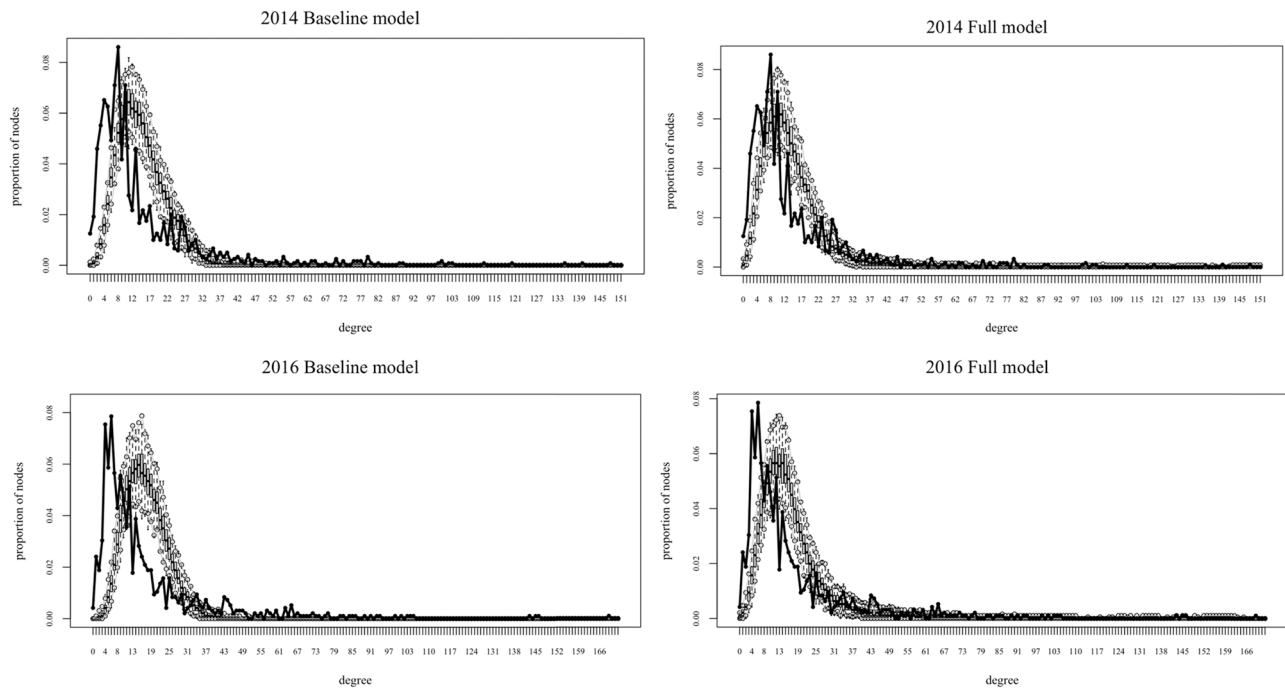


Fig. A2. (continued).

illustrates that the shares of well-approximated statistics started to differ noticeably from 1993 onward.

Second, simulated degree distributions from full models exhibit a higher skewness than the simulated degree distributions stemming from baseline models. This trend is especially visible during the last five conferences and further corroborates the notion that researchers' characteristics affected their prominence as collaborators. It also strengthens our interpretation of Fig. 3 in the main text, which shows that full models perform better in approximating the empirical skewness of the degree distribution.

Third, Fig. A2 shows that full models increase the fit for nodes with a high degree. This trend is visible from 1996 onward: while the boxplots remain "flat" for high-degree nodes in baseline models, the range of simulated values is extended in full models, which is indicated by the empty dots representing the upper end of confidence intervals in this plot type. In the long tail of the degree distribution, simulated values of full models show a wider confidence interval and thereby cover empirical values more often than baseline models.

In summary, our additional GOF analyses for the degree distribution further support our substantial claims. Models including researchers' characteristics—i.e., years of experience, publications, and last author positions—improve the fit of the degree distribution, especially during the second half of the field's development, in comparison to baseline models. This points to an increased relevance of accumulation dynamics as the field grew and matured.

Trends in inequality and simulated values from network models

Here we report similar analyses as performed in Fig. 2 in the main text. The only difference is that we used the standard deviation and the Gini coefficient as alternative measures for inequality in the network's degree distribution (Badham, 2013; Snijders and Steglich, 2015).

The results presented in Fig. A3 are in line with our results reported in the main text: inequality in empirical networks increased over time, and models including actor attributes simulated values that are closer to empirical values than simulated values from baseline models.

Results for the Gini coefficient are summarized in Fig. A4 and did not indicate a strong increase in inequality. Moreover, while full models were closer to simulating empirical values than baseline models, these differences were less pronounced in comparison with other measures for inequality. These findings are probably due to the fact that the Gini coefficient places equal emphasis on all percentiles of a distribution, whereas the skewness and the standard deviation are more sensitive to the top ranks of a distribution. While the Gini coefficient considers that most researchers did not have many collaborative ties from the outset of the field, the standard deviation and the skewness capture the advent of authors accumulating very large numbers of collaboration partners. Given that we are interested in the formation of elites rather than the uniform distribution of the total number of coauthor ties, we believe that the skewness and the standard deviation are better measures for our purpose (Tables A2–A4).

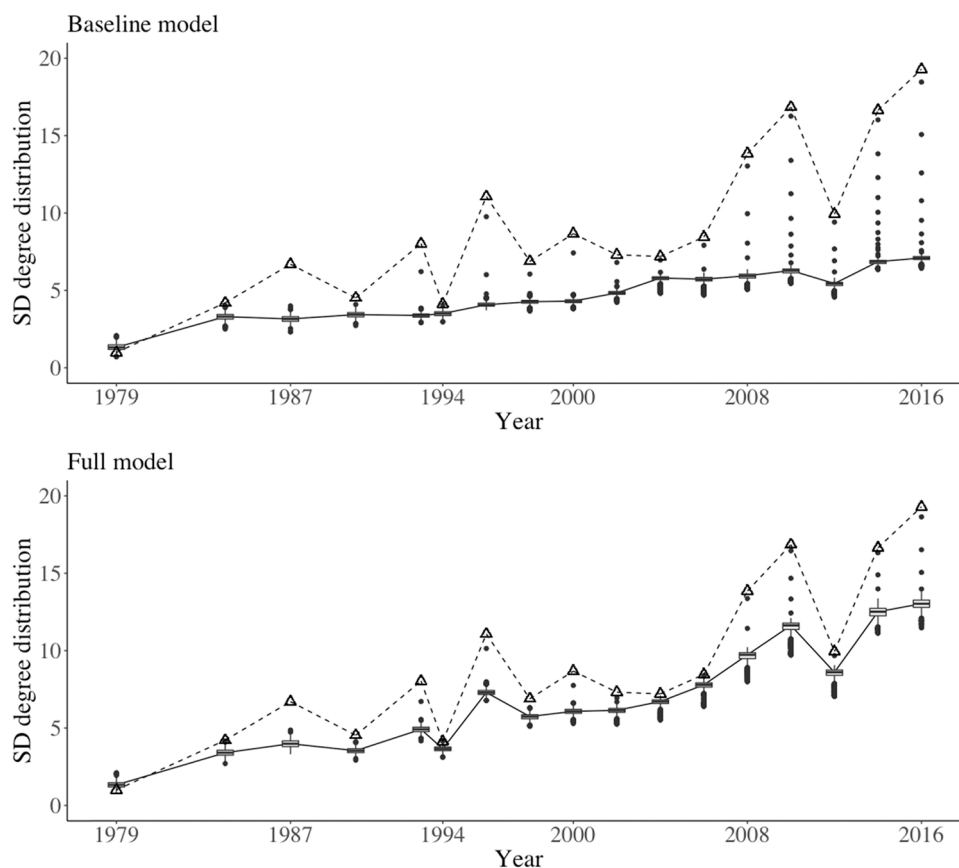


Fig. A3. Inequality in coauthor ties over time measured by the standard deviation. Note.—Each box plot represents the distribution of standard deviations calculated in simulated networks obtained from network models reported in [Tables 3 and 4](#). Boxplots in the upper panel show simulated values for networks generated according to our baseline model specification (see [Table 3](#)). The bottom panel reports simulated values generated from the full models that additionally include researchers' characteristics ([Table 4](#)). We simulated 1,000 networks for each year and specification. Triangles indicate empirical values, i.e., the observed standard deviation of the degree distribution in a particular year. The dashed line connects empirical values, while the straight line follows the medians of simulated values.

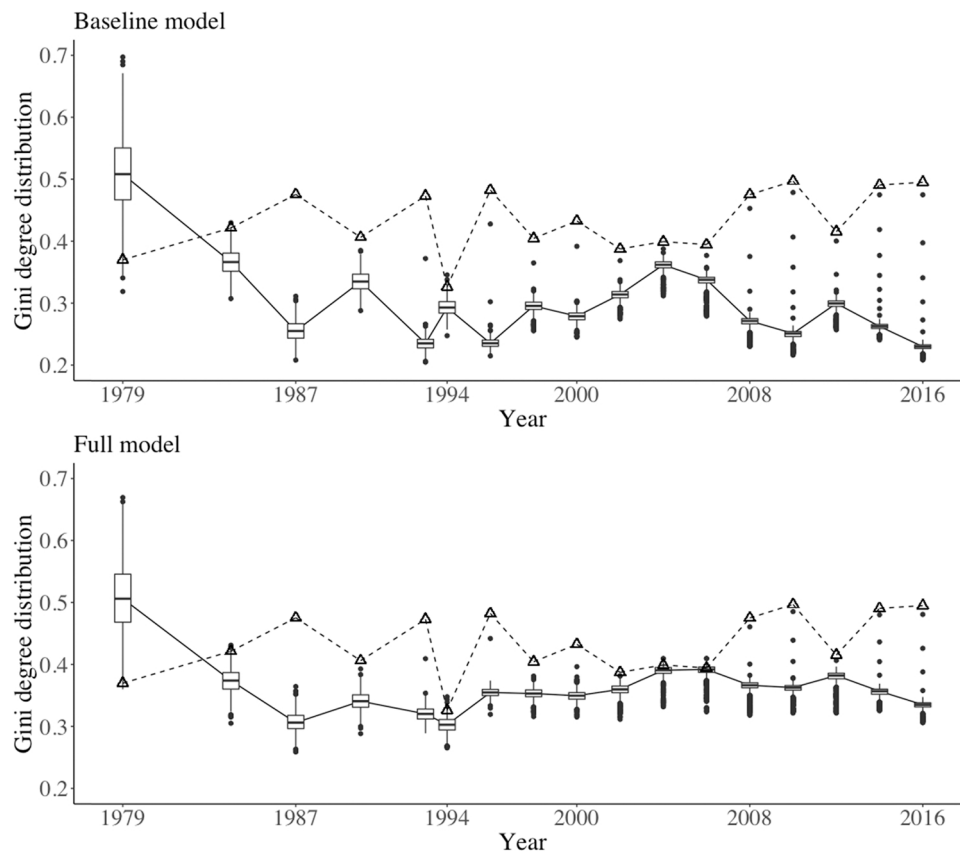


Fig. A4. Inequality in coauthor ties over time measured by the Gini coefficient. Note.—Each box plot represents the distribution of Gini coefficients calculated in simulated networks obtained from network models reported in Tables 3 and 4. Boxplots in the upper panel show simulated values for networks generated according to our baseline model specification (see Table 3). The bottom panel reports simulated values generated from the full models that additionally include researchers' characteristics (Table 4). We simulated 1,000 networks for each year and specification. Triangles indicate empirical values, i.e., the observed Gini coefficient of the degree distribution in a particular year. The dashed line connects empirical values, while the straight line follows the medians of simulated values.

Table A2

The role of experience for the collaboration network of neuroblastoma researchers in exponential random graph models (ERGMs).

Years	1979		1984		1987		1990		1993		1994	
	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	-0.003 (0.021)	-0.076	0.094*** (0.008)	1.801	0.054*** (0.006)	0.874	0.074*** (0.003)	2.241	0.046*** (0.002)	1.481	0.079*** (0.003)	2.411
Same institution	0.108*** (0.017)	2.709	0.092*** (0.008)	1.755	0.096*** (0.009)	1.559	0.053*** (0.003)	1.605	0.058*** (0.002)	1.881	0.031*** (0.003)	0.937
Popularity years of experience	0.001 (0.006)	0.034	-0.004 (0.003)	-0.086	0.011*** (0.002)	0.174	0.040*** (0.001)	0.119	0.010*** (0.001)	0.309	0.005*** (0.001)	0.155
Difference in authors' years of experience	-0.009 (0.008)	-0.227	0.001 (0.004)	0.011	-0.005 (0.003)	-0.080	-0.004* (0.002)	-0.123	-0.005*** (0.001)	-0.173	-0.005*** (0.001)	-0.162
Baseline probability	0.040		0.053		0.062		0.033		0.031		0.032	
Years	1996		1998		2000		2002		2004		2006	
	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	0.037*** (0.001)	1.807	0.033*** (0.001)	2.083	0.030*** (0.001)	1.649	0.030*** (0.001)	2.213	0.021*** (0.0004)	2.301	0.020*** (0.0004)	1.893
Same institution	0.035*** (0.001)	1.677	0.027*** (0.001)	1.701	0.035*** (0.001)	1.965	0.025*** (0.0005)	1.875	0.019*** (0.0003)	2.089	0.023*** (0.0003)	2.251
Popularity years of experience	0.006*** (0.0003)	0.267	0.005*** (0.0002)	0.290	0.006*** (0.0002)	0.328	0.004*** (0.0002)	0.299	0.002*** (0.0001)	0.223	0.003*** (0.0001)	0.317
Difference in authors' years of experience	-0.004*** (0.0004)	-0.213	-0.005*** (0.0004)	-0.314	-0.005*** (0.0004)	-0.259	-0.003*** (0.0003)	-0.198	-0.001*** (0.0002)	-0.125	-0.002*** (0.0002)	-0.166
Baseline probability	0.021		0.016		0.018		0.013		0.009		0.010	
Years	2008		2010		2012		2014		2016			
	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	0.021*** (0.0004)	1.647	0.020*** (0.0003)	1.672	0.017*** (0.0003)	1.934	0.022*** (0.0003)	1.824	0.029*** (0.0004)	1.578		
Same institution	0.026*** (0.0004)	2.078	0.026*** (0.0004)	2.166	0.021*** (0.0003)	2.286	0.026*** (0.0003)	2.144	0.037*** (0.0005)	2.066		
Popularity years of experience	0.005*** (0.0001)	0.361	0.005*** (0.0001)	0.393	0.003*** (0.0001)	0.309	0.004*** (0.0001)	0.363	0.007*** (0.0001)	0.364		
Difference in authors' years of experience	-0.009*** (0.0002)	-0.148	-0.002*** (0.0002)	-0.200	-0.001*** (0.0001)	-0.082	-0.001*** (0.0001)	-0.114	-0.002*** (0.0002)	-0.133		
Baseline probability	0.013		0.011		0.009		0.012		0.018			

Note.—All continuous variables are z-standardized to enhance the comparability of estimates across models. Delta standard errors (Duxbury, 2019) are reported in parentheses. Scaled AMEs are AMEs divided by the baseline probability and can be interpreted as relative changes in tie probability if a network variable increases by one unit. We multiplied scaled AMEs by 100 to provide a measure capturing the percentage change of the baseline probability in Fig. 3 and Fig. 4.

† p < 0.10 * p < 0.05 ** p < 0.01 *** p < 0.001 (two-sided)

Table A3

The role of productivity for the collaboration network of neuroblastoma researchers in exponential random graph models (ERGMs).

Years	1979		1984		1987		1990		1993		1994	
	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	-0.005 (0.021)	-0.130	0.095*** (0.008)	1.807	0.065*** (0.006)	1.059	0.074*** (0.003)	2.258	0.046*** (0.002)	1.500	0.078*** (0.003)	2.385
Same institution	0.107*** (0.017)	2.684	0.090*** (0.008)	1.715	0.092*** (0.009)	1.488	0.055*** (0.003)	1.658	0.060*** (0.002)	1.920	0.033*** (0.003)	1.013
Popularity cumulated publications	0.009 (0.010)	0.226	-0.0004 (0.006)	-0.002	0.015*** (0.002)	0.238	0.006*** (0.001)	0.186	0.011*** (0.001)	0.343	0.006*** (0.001)	0.173
Difference cumulated publications	-0.025 (0.016)	-0.618	0.006 (0.006)	0.116	0.005 (0.003)	0.087	0.001 (0.002)	0.031	-0.004*** (0.001)	-0.123	-0.003* (0.0015)	-0.095
Baseline probability	0.040		0.053		0.062		0.033		0.031		0.032	
Years	1996		1998		2000		2002		2004		2006	
	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	0.037*** (0.001)	1.796	0.032*** (0.001)	2.039	0.029*** (0.001)	1.608	0.029*** (0.001)	1.608	0.021*** (0.0004)	2.285	0.019*** (0.0004)	1.841
Same institution	0.036*** (0.001)	1.751	0.028*** (0.001)	1.793	0.036*** (0.001)	2.011	0.026*** (0.0005)	2.011	0.020*** (0.0003)	2.102	0.024*** (0.0003)	2.289
Popularity cumulated publications	0.009*** (0.0003)	0.433	0.006*** (0.0002)	0.361	0.007*** (0.0002)	0.386	0.005*** (0.0002)	0.386	0.002*** (0.0001)	0.256	0.004*** (0.0001)	0.377
Difference cumulated publications	-0.003*** (0.0004)	-0.136	-0.002*** (0.0003)	-0.117	-0.003*** (0.0004)	-0.179	-0.002*** (0.0003)	-0.179	-0.004*** (0.0002)	-0.073	-0.002*** (0.0002)	-0.165
Baseline probability	0.021		0.016		0.018		0.013		0.009		0.010	
Years	2008		2010		2012		2014		2016			
	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	0.020*** (0.0004)	1.622	0.019*** (0.0003)	1.640	0.017*** (0.0003)	1.895	0.022*** (0.0003)	1.790	0.028*** (0.0004)	1.563		
Same institution	0.026*** (0.0004)	2.088	0.025*** (0.0003)	2.158	0.021*** (0.0003)	2.305	0.026*** (0.0003)	2.125	0.036*** (0.0005)	2.013		
Popularity cumulated publications	0.005*** (0.0001)	0.422	0.006*** (0.0001)	0.469	0.003*** (0.0001)	0.380	0.005*** (0.0001)	0.438	0.008*** (0.0001)	0.438		
Difference cumulated publications	-0.002*** (0.0001)	-0.181	-0.003*** (0.0001)	-0.243	-0.001*** (0.0001)	-0.146	-0.002*** (0.0001)	-0.191	-0.004*** (0.0002)	-0.208		
Baseline probability	0.013		0.011		0.009		0.012		0.018			

Note.—All continuous variables are z-standardized to enhance the comparability of estimates across models. Delta standard errors (Duxbury, 2019) are reported in parentheses. Scaled AMEs are AMEs divided by the baseline probability and can be interpreted as relative changes in tie probability if a network variable increases by one unit. We multiplied scaled AMEs by 100 to provide a measure capturing the percentage change of the baseline probability in Fig. 3 and Fig. 4.

† $p < 0.10$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$ (two-sided)

Table A4

The role of last authorships for the collaboration network of neuroblastoma researchers in exponential random graph models (ERGMs).

Years	1979		1984		1987		1990		1993		1994	
	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	-0.004 (0.021)	-0.092	0.094*** (0.008)	1.791	0.064*** (0.006)	1.038	0.075*** (0.003)	2.280	0.049*** (0.002)	1.594	0.080*** (0.003)	2.432
Same institution	0.109*** (0.017)	2.741	0.092*** (0.008)	1.740	0.095*** (0.009)	1.538	0.052*** (0.003)	1.591	0.055*** (0.003)	1.769	0.031*** (0.003)	0.941
Popularity share last author positions	0.0004 (0.007)	0.004	-0.007 (0.006)	-0.139	-0.009 (0.005)	-0.151	-0.013*** (0.003)	-0.385	-0.0025* (0.0012)	-0.080	-0.007*** (0.002)	-0.208
Difference share last author positions	0.002 (0.009)	0.052	0.009 (0.006)	0.174	0.003 (0.006)	0.046	0.011** (0.003)	0.318	0.0001 (0.001)	0.003	0.006** (0.002)	0.194
Baseline probability	0.040		0.053		0.062		0.033		0.031		0.032	
Years	1996		1998		2000		2002		2004		2006	
	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	0.039*** (0.001)	1.895	0.034*** (0.001)	2.119	0.030*** (0.001)	1.676	0.031*** (0.001)	2.269	0.022*** (0.0004)	2.348	0.020*** (0.0004)	1.909
Same institution	0.034*** (0.001)	1.658	0.027*** (0.001)	1.725	0.035*** (0.001)	1.981	0.025*** (0.001)	1.890	0.020*** (0.0003)	2.108	0.024*** (0.0003)	2.295
Popularity share last author positions	0.0009 (0.0005)	0.044	0.0004 (0.0004)	0.026	0.0010* (0.0005)	0.056	0.002*** (0.0003)	0.115	0.0004* (0.0002)	0.043	0.001*** (0.0002)	0.067
Difference share last author positions	-0.001* (0.0006)	-0.066	0.0005 (0.0005)	0.034	0.0006 (0.0006)	0.032	>0.0001 (0.0003)	-0.001	0.0002 (0.0002)	0.020	-0.0002 (0.0002)	-0.019
Baseline probability	0.021		0.016		0.018		0.013		0.009		0.010	

(continued on next page)

Table A4 (continued)

Years	2008		2010		2012		2014		2016	
	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	0.021*** (0.0004)	1.701	0.021*** (0.0003)	1.777	0.018*** (0.0003)	2.006	0.023*** (0.0003)	1.877	0.029*** (0.0004)	1.630
Same institution	0.026*** (0.0004)	2.059	0.025*** (0.0003)	2.090	0.021*** (0.0003)	2.309	0.026*** (0.0003)	2.106	0.036*** (0.0005)	1.981
Popularity share last author positions	0.001*** (0.0002)	0.106	0.002*** (0.0001)	0.144	0.001*** (0.0001)	0.108	0.002*** (0.0001)	0.146	0.001*** (0.0002)	0.077
Difference share last author positions	-0.001*** (0.0002)	-0.090	-0.001*** (0.0002)	-0.123	-0.0004* (0.0002)	-0.048	-0.001*** (0.0002)	-0.095	-0.002*** (0.0003)	-0.102
Baseline probability	0.013		0.011		0.009		0.012		0.018	

Note.—All continuous variables are z-standardized to enhance the comparability of estimates across models. Delta standard errors (Duxbury, 2019) are reported in parentheses. Scaled AMEs are AMEs divided by the baseline probability and can be interpreted as relative changes in tie probability if a network variable increases by one unit. We multiplied scaled AMEs by 100 to provide a measure capturing the percentage change of the baseline probability in Fig. 3 and Fig. 4.
† p < 0.10 * p < 0.05 ** p < 0.01 *** p < 0.001 (two-sided)

Online Appendix B. Supporting information

Further analysis associated with this article can be found in the online version at doi:10.1016/j.socnet.2022.05.001.

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